

Strikingly different neurotransmitter release strategies in dopaminergic subclasses

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This study provides evidence for distinct neurotransmitter release modalities between two subclasses of dopaminergic neurons in the olfactory bulb. Specifically, it demonstrates dendritic neurotransmitter release in anaxonic neurons and axonal release in axon-bearing neurons. The presence of GABAergic self-inhibition in anaxonic neurons further underscores the functional divergence between these subtypes. Overall, the manuscript presents **solid** evidence and offers biologically **important** insights into the organization and function of dopaminergic circuits within the olfactory bulb.

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Abstract

Neuronal function is intimately tied to axodendritic polarity. Neurotransmitter release, for example, is usually the role of the axon. There are widespread exceptions to this rule, however, including many mammalian neuronal types that can release neurotransmitter from their dendrites. In the mouse olfactory bulb, closely related subclasses of dopaminergic interneuron differ markedly in their polarity, with one subtype lacking an axon entirely. These axon-bearing and anaxonic dopaminergic subclasses have distinct developmental profiles and sensory responses, but how their fundamental polarity differences translate to functional outputs remains entirely unknown. Here, we provide anatomical evidence for distinct neurotransmitter release strategies among these closely related dopaminergic subtypes: anaxonic cells release from their dendrites, while axon-bearing neurons release exclusively from their intermittently myelinated axon. These structural differences are linked to a clear functional distinction: anaxonic, but not axon-bearing dopaminergic neurons are capable of self-inhibition. Our findings suggest that variations in polarity can produce striking distinctions in neuronal outputs, and that even closely related neuronal subclasses may play entirely separate roles in sensory information processing.

Introduction

Neurons are one of the most striking examples of cell polarisation. Most neurons have canonical axodendritic polarity, defined by the presence of a somatodendritic domain and a single axon (Banker, 2018 [🔗](#)). The morphological and molecular specialisation of these distinct compartments gives them unique functional properties, with dendrites usually receiving most synaptic inputs, and axons sending outputs to other neurons by releasing neurotransmitters. However, exceptions to this neuronal polarity dogma are common. In the mammalian brain, there are neurons – including midbrain dopaminergic cells (Pucak and Grace, 1994 [🔗](#); Falkenburger et al., 2001 [🔗](#)) and cortical bitufted cells (Zilberter et al., 1999 [🔗](#)) – that release neurotransmitter from their dendrites. There are also neurons – including retinal amacrine cells (Wu et al., 2011 [🔗](#); Goillard et al., 2020 [🔗](#)) – that lack canonical axons altogether. A remarkable hub for such unconventional neuronal polarity is the olfactory bulb (OB), the first part of the brain that processes information about the sense of smell. Dendritic release of neurotransmitters is a widespread phenomenon in the OB, even in projection neurons and other cells with classically defined axodendritic polarity (Schoppa and Urban, 2003 [🔗](#); Hayar et al., 2004 [🔗](#); Imamura et al., 2020 [🔗](#)). In addition, non-canonical polarity is extremely common in the OB, where almost all GABAergic interneurons lack an axon entirely (Tufo et al., 2022 [🔗](#)).

Neuronal polarity in the OB can be completely different even in closely related neuronal subtypes. In the glomerular layer, a major class of interneurons – around 90,000 cells total in the mouse OB (McLean and Shipley, 1988 [🔗](#); Panzanelli et al., 2007 [🔗](#); Parrish-Aungst et al., 2007 [🔗](#)) – releases both dopamine and GABA in order to modulate neurotransmitter release from olfactory sensory neuron terminals and to regulate downstream processing in local circuits (Wachowiak and Cohen, 1999 [🔗](#); Berkowicz and Trombley, 2000 [🔗](#); Borisovska et al., 2013 [🔗](#); Liu et al., 2013 [🔗](#); Banerjee et al., 2015 [🔗](#); Vaaga et al., 2017 [🔗](#); Garg et al., 2025 [🔗](#)). We recently found that these dopaminergic (DA) neurons can be classified into two subtypes depending on a binary morphological distinction: the presence or absence of an axon. This fundamental divergence in polarity in anaxonic and axon-bearing DA cells is correlated with different excitability profiles and odour response properties (Chand et al., 2015 [🔗](#); Galliano et al., 2018 [🔗](#); Lau et al., 2024 [🔗](#)). It also corresponds closely to previous descriptions of OB DA heterogeneity based on cell size (Kosaka and Kosaka, 2008 [🔗](#); Pignatelli and Belluzzi, 2017 [🔗](#)), with axon-bearing DA neurons having significantly larger somas (Kosaka and Kosaka, 2008 [🔗](#); Galliano et al., 2018 [🔗](#)). Whilst axon-bearing DA neurons represent the minority in terms of numbers, making up ~2.5 % of the OB's DA population (Galliano et al., 2018 [🔗](#)), they have the potential to significantly influence olfactory processing via inter-glomerular inhibition (Kosaka & Kosaka, 2008 [🔗](#); Liu et al., 2013 [🔗](#); Whitesell et al., 2013 [🔗](#); Banerjee et al., 2015 [🔗](#)). However, the specific function of these two neuron subtypes within OB glomerular networks is not fully understood. A crucial outstanding question is how the structural differences in these two neuron subtypes translate into strategies for influencing bulbar circuits via neurotransmitter release.

Anaxonic DA neurons only have a soma and dendrites (Chand et al., 2015 [🔗](#); Galliano et al., 2018 [🔗](#)) making these the sole available domains for neurotransmitter release. Indeed, dendritic presynaptic structures are present in DA neurons (Kiyokage et al., 2017 [🔗](#)), and are associated with these cells' functional ability to self-inhibit via local GABA release (Smith and Jahr, 2002 [🔗](#); Murphy et al., 2005 [🔗](#); Maher and Westbrook, 2008 [🔗](#)). On the other hand, axon-bearing DA neurons are known to be the sole source of long-range interglomerular connections in the OB (Kosaka and Kosaka, 2008 [🔗](#)), with functional studies showing that these cells release neurotransmitters over these long-distance axonal projections (Liu et al., 2013 [🔗](#); Whitesell et al., 2013 [🔗](#); Banerjee et al., 2015 [🔗](#)). However, in a brain region where dendritic release is the norm rather than the exception, what remains unknown is whether axon-bearing DAs combine this

distal, *interglomerular* signalling with local, *intraglomerular* inhibition via release from their dendrites. Answering this question has important implications for how information is processed in early olfactory circuits.

Here, we provide anatomical evidence showing that axon-bearing OB DA neurons have axons with neurotransmitter release sites, while their dendrites lack these sites almost entirely. On the other hand, we found that all anaxonic DA neurons have release sites on their dendrites. These structural differences are associated with a striking functional distinction: while all anaxonic neurons can self-inhibit, there is an almost total absence of self-inhibition responses in the axon-bearing subpopulation. This polarity-based dichotomy suggests that even these closely related interneuron subtypes may play entirely separate roles in the processing of sensory information.

Results

Labelling putative presynaptic structures in individual olfactory bulb dopaminergic neurons

We first developed a structural approach for labelling putative presynaptic sites in individual OB DA neurons. We observed that synaptophysin, a membrane protein located in vesicles at the presynapse, is expressed in OB DA cells at both the mRNA (**Figure 1A**) and protein (**Figure 1B**) levels. Given the high synaptic density and complexity of OB glomeruli, it is not possible to reliably assign specific synaptophysin label to individual DA neurons using widespread staining of endogenous protein. We therefore chose a sparse labelling approach based on the injection of an AAV-DJ-hSyn-FLEX-mGFP-T2A-Synaptophysin-mRuby construct (Beier et al., 2015; Kathe, et al., 2022). This vector allows the expression in individual cre-expressing cells of a membrane-bound GFP for cell morphology, and fluorophore-tagged full-length synaptophysin for the visualisation of putative presynaptic puncta. There is strong evidence showing that the overexpression of fluorophore-tagged synaptophysin is functionally benign (Granseth et al., 2006; Li and Tsien, 2012).

We performed *in utero* injections in embryonic day 14 (E14) vGAT (vesicular GABA transporter)-cre mice. Since OB dopaminergic neurons are also GABAergic (Kosaka and Kosaka 2008; 2016; Maher & Westbrook 2008; Borisovska et al., 2013; Vaaga et al., 2016; Liu et al. 2013) and express vGAT at both the mRNA (**Figure 1A**) and protein (**Figure 1B**; Panzanelli et al., 2007) levels, this conditional line gives access to both anaxonic and axon-bearing DA subclasses, alongside many non-dopaminergic OB GABA-releasing neurons (Galliano et al., 2018). Dopaminergic neurons were then specifically identified via immunostaining against tyrosine hydroxylase (TH). TH is the rate-limiting enzyme in catecholamine synthesis and, because of the absence of noradrenergic neurons in the OB, is a specific and comprehensive marker for all dopaminergic cells in this brain region (Hokfelt et al., 1975; Rosser et al., 1986; Kosaka and Kosaka 2016; Galliano et al., 2018).

We chose an embryonic timepoint for viral targeting to ensure the effective and sparse labelling of axon-bearing and anaxonic DA neurons (**Figure 1C**). Early infection before cells have migrated to their final positions produced sparse labelling which, when analysed in juvenile postnatal animals, permitted the unambiguous identification of an individual neuron's processes. Embryonic targeting also had the benefit of equally sampling DA subclasses. While anaxonic DA neurons are born throughout life, axon-bearing DA cells are exclusively generated in embryonic development (Batista-Brito et al., 2008; Galliano et al., 2018). Embryonic injections therefore ensured effective and more balanced labelling of both populations.

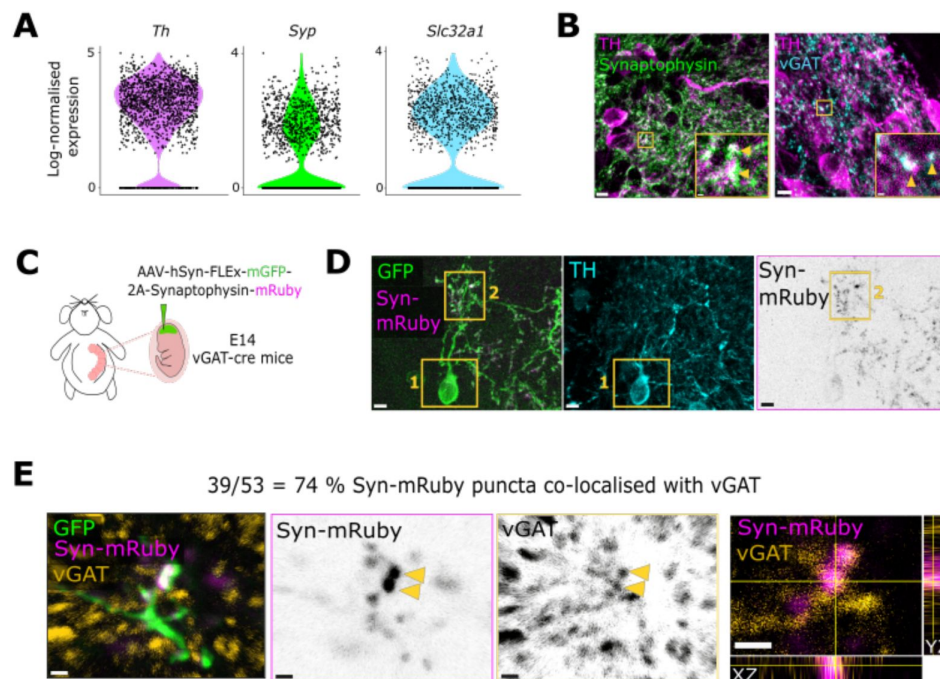


Figure 1.

Labelling putative presynaptic structures in individual olfactory bulb dopaminergic neurons

(A) Log₂-normalised expression of *Th* (tyrosine hydroxylase), *Syp* (Synaptophysin) and *Slc32a1* (vGAT) mRNA in DA neurons from single cell RNA sequencing data (Brann et al., 2020). **(B)** Example confocal images of endogenous immunostaining for synaptophysin (green) and TH (magenta) on the left, and vGAT (blue) and TH (magenta) on the right. Both images were taken in the glomerular layer of the OB. Yellow arrowheads point to small clusters where TH and synaptophysin or TH and vGAT co-localise. Scalebars: 5 μ m (main) and 0.5 μ m (inset) for the images on the left, 4 μ m (main) and 1 μ m (inset) for the images on the right. **(C)** Strategy to label putative presynaptic release sites in individual DA neurons. **(D)** Example confocal image of a successfully labelled DA cell. Inset 1 reveals the TH+ dopaminergic identity of the neuron (cyan) and inset 2 highlights the Syn-mRuby puncta (magenta, black). Scalebars, 5 μ m. **(E)** Example confocal image of a GFP+ (green), Syn-mRuby+ (magenta, black) neuronal process co-stained with vGAT (orange, black). Yellow arrowheads indicate examples where Syn-mRuby and vGAT puncta co-localise. The last panel shows the orthogonal views of the bottom punctum. Scalebars, 1 μ m.

We were able to detect well-defined synaptophysin-mRuby puncta along the processes of TH+ GFP+ neurons (**Figure 1D** [↗](#)). To assess whether this method labels reliable presynaptic release sites in DA neurons, we co-stained the injected tissue against vGAT and observed high co-localisation between endogenous vGAT and mRuby label (**Figure 1E** [↗](#); 39/53 = 74 % synaptophysin-mRuby puncta co-localised with vGAT; n = 9 cells from N = 4 mice). These results demonstrate that a large proportion of synaptophysin-mRuby puncta are likely to reflect putative presynaptic release sites in OB DA neurons.

Anaxonic DA neurons have dendritic neurotransmitter release sites

Given that anaxonic DA neurons lack an axon, and are known to release both GABA and dopamine, to be able to release neurotransmitter they must have presynaptic release sites on their dendrites. We identified this neuronal subtype in our population of GFP+/TH+ neurons by the lack of the proximal axonal marker TRIM46 (**Figure 2A** [↗](#), [van Beuningen et al., 2015](#) [↗](#); [Galliano et al., 2018](#) [↗](#)).

As predicted, we found that all anaxonic neurons had synaptophysin-mRuby puncta in their dendrites (**Figure 2B, C** [↗](#); n=9 anaxonic DA cells from N=4 mice). The thin slices necessary for puncta detection precluded any comprehensive analysis of dendritic morphology or synaptic distributions. The resultant sub-sampling of individual cells' arbours may also have contributed to the variability in puncta density we observed across neurons (**Figure 2C** [↗](#)), although similarly high variability between OB dopaminergic cell presynaptic specialisations has been shown in previous ultrastructural analyses ([Kiyokage et al., 2017](#) [↗](#)). We saw no consistent evidence for spatial patterns, clusters, or privileged branches; instead, puncta were distributed throughout each cell's dendritic arbour (**Figure 2C** [↗](#)). These results provide anatomical evidence for the presence of dendritic neurotransmitter release sites in anaxonic DA neurons.

Axon-bearing DA neurons have release sites on their intermittently myelinated axons, but not on their dendrites

Long-range GABA release has been functionally demonstrated in dopaminergic neurons ([Liu et al., 2013](#) [↗](#); [Whitesell et al., 2013](#) [↗](#); [Banerjee et al., 2015](#) [↗](#)), strongly suggesting that axon-bearing DA neurons have axonal release sites. It was not possible to follow TRIM46+ DA axons for long distances from the soma in thin brain slices, so instead we used the distal axon marker myelin basic protein (MBP). In the first description of myelination in this cell type, we found that DA axons are myelinated in an intermittent pattern, a common trait in other brain regions (**Figure 3A** [↗](#), [Tomassy et al., 2014](#) [↗](#)). Using MBP as indicator of axon identity, we were then able to locate synaptophysin-mRuby puncta in unmyelinated sections of GFP+ TH+ axons (**Figure 3B** [↗](#) and **Supplementary Figure 1** [↗](#); n=5 axon-bearing DA cells from N=3 mice). As predicted, axon-bearing DA neurons therefore have putative presynaptic release sites on their axons.

Most neurons with canonical axodendritic polarity in the olfactory bulb release neurotransmitters from both axons and dendrites ([Castro and Urban, 2009](#) [↗](#)). To assess if this scenario also holds true for DA axon-bearing cells, we explored the dendritic localisation of synaptophysin-mRuby-labelled presynaptic release sites in GFP+ neurons with clear TRIM46+ proximal axonal domains (**Figure 3C** [↗](#)). As expected, these axon-bearing OB DA cells had larger somas than their anaxonic neighbours (**Figure 3D** [↗](#) and **Supplementary Figure 2A** [↗](#); soma area: $110.3 \pm 7.72 \mu\text{m}^2$ for axon-bearing DA cells, $61.99 \pm 4.41 \mu\text{m}^2$ for anaxonic DA neurons; n = 11 axon-bearing cells and n = 9 anaxonic neurons from N = 5 mice; unpaired t-test with Welch's correction, $p < 0.0001$). Given previous findings that axon-bearing DA cells have more widely ramified dendritic trees ([Galliano et al., 2018](#) [↗](#)), we ensured that our reconstructions of TRIM46-negative dendritic processes were equally extensive in both cell subtypes (**Supplementary Figure 2B** [↗](#); maximum length of reconstructed dendrites: $67.64 \pm 8.96 \mu\text{m}$ for axon-bearing DA cells, $74.88 \pm 11.68 \mu\text{m}$ for anaxonic

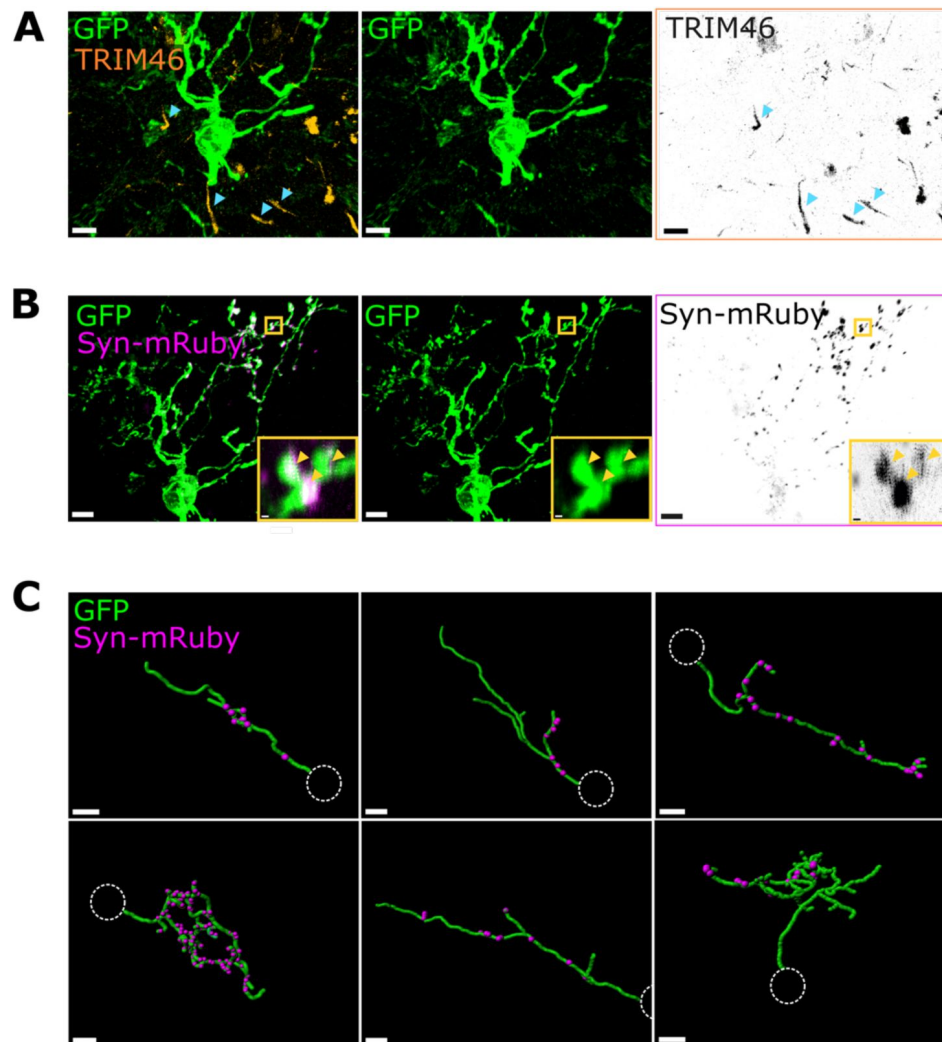


Figure 2.

Anaxonic DA neurons have dendritic neurotransmitter release sites

(A) Example confocal image of a TRIM46-negative anaxonic DA neuron. Blue arrowheads point to examples of other TRIM46+ AISs (orange, black) in the same region which do not co-localise with this neuron's GFP signal. Scalebars, 5 μ m. **(B)** Snapshot of the same neuron in (A) showing Synaptophysin-mRuby Puncta (magenta, black) on the dendrites. Yellow inset highlights a region of the neuron with multiple mRuby+ puncta within the GFP+ (green) processes (yellow arrows). Scalebars, 5 μ m and 0.5 μ m. **(C)** Example snapshots from 3D dendritic reconstructions (green, GFP) and presynaptic puncta detection (magenta, Syn-mRuby) of anaxonic DA neurons. Note: these are not full dendritic reconstructions, but example dendrites. Dotted white circle represents the soma location. Scalebars, 5 μ m.

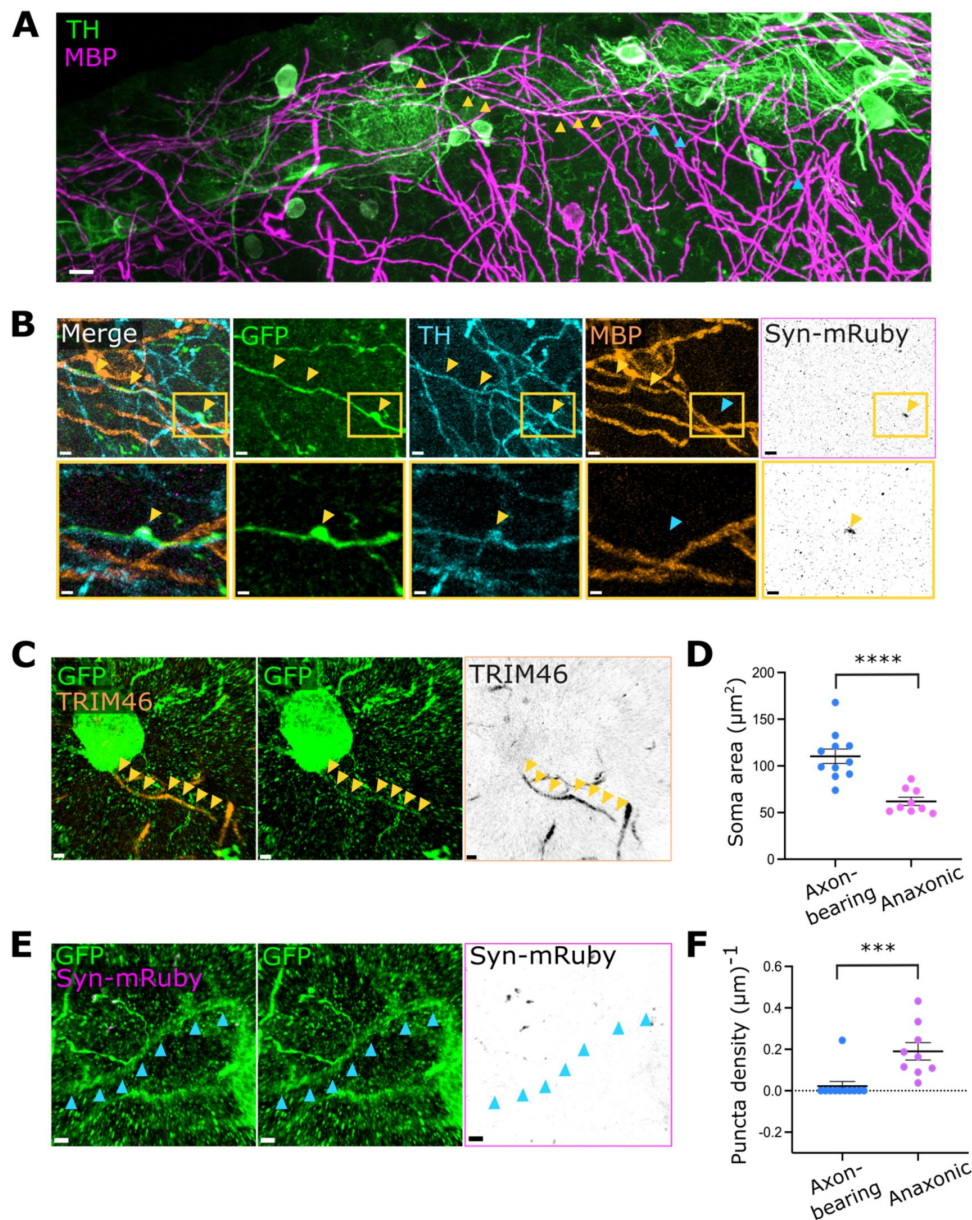


Figure 3.

Axon-bearing DA neurons have release sites on their intermittently myelinated axons, but not on their dendrites

(A) Stitching of individual confocal stacks processed for maximum intensity projections of OB DA neurons co-stained with TH (green) and myelin basic protein (MBP, magenta). Yellow arrowheads point to myelinated parts of the axon, blue arrowheads show unmyelinated areas. Scalebar, 10 μm . (B) Example confocal images of a distal DA axon stained with GFP (green), TH (cyan), MBP (orange) and synaptophysin-mRuby (magenta, black). Yellow inset highlights the location of the presynaptic bouton. Yellow arrowheads point to co-localised regions, blue arrowheads show non-co-localisation. Scalebars, 2 μm and 1 μm . (C) Confocal image of an axon-bearing TRIM46+ DA neuron. Yellow arrowheads show co-localised staining for GFP (green) and TRIM46 (orange, black). Scalebars, 2 μm . (D) Soma area of axon-bearing and anaxonic DA neurons. Each dot shows one cell; lines show mean $\pm$ SEM; $n = 11$ axon-bearing cells and $n = 9$ anaxonic neurons from $N = 5$ mice; unpaired t-test with Welch's correction; ****, $p < 0.0001$. (E) Snapshot of the same axon-bearing DA neuron shown in (C), co-stained with GFP (green) and synaptophysin-mRuby (magenta, black). Blue arrows show dendritic segments lacking mRuby label, despite the presence of clear mRuby+ puncta in neighbouring processes from a different GFP+ cell. Scalebars, 2 μm . (F) Dendritic puncta density in axon-bearing and anaxonic DA neurons. All conventions as in D; $n = 11$ axon-bearing cells and $n = 9$ anaxonic neurons from $N = 5$ mice; Mann-Whitney test; **, $p = 0.0001$.

DA neurons; $n = 11$ axon-bearing cells and $n = 9$ anaxonic neurons from $N = 5$ mice; unpaired t-test with Welch's correction, $p = 0.6299$). We found that, unlike their anaxonic counterparts, almost all TRIM46+ neurons (10/11 cells from $N = 4$ mice, = 90.9 %) did not have any synaptophysin-mRuby puncta on their dendrites (**Figure 3E, F**); dendritic puncta density: $0.02 \pm 0.02 (\mu\text{m})^{-1}$ for axon-bearing cells, $0.19 \pm 0.04 (\mu\text{m})^{-1}$ for anaxonic neurons; $n = 11$ axon-bearing cells and $n = 9$ anaxonic neurons from $N = 5$ mice; Mann-Whitney test; $p = 0.0001$). We did find one single axon-bearing DA neuron with apparent dendritic puncta; however, this cell had dramatically higher levels and unusual patterns of expression as compared to all other GFP+ neurons (**Supplementary Figure 2C, D**) and we include it in our sample only through an abundance of caution. These findings provide a structural basis for dendritic neurotransmitter release in anaxonic DA neurons, but not in the axon-bearing population of DA neurons in the OB.

Anaxonic DA neurons are capable of GABAergic self-inhibition

The fact that dendritic release sites are likely located near dendritic receptors (Kiyokage et al., 2017) might have the functional consequence that anaxonic DA cells can signal to themselves and influence their own activity. We found that OB DA neurons do not express DA receptors at the mRNA level (**Figure 4A**), making it unlikely that they could influence their own activity via DA signalling. However, GABAergic auto-evoked inhibition (AEI) or self-inhibition has been described in OB periglomerular cells in general, and in DA neurons in particular (Smith and Jahr, 2002; Murphy et al., 2005; Maher and Westbrook, 2008; Borisovska et al., 2013).

To address if the anaxonic subtype of DA neuron can self-inhibit, we used the DAT-tdT mouse line to specifically target DA neurons (Bäckman et al., 2006; Madisen et al., 2009; Galliano et al., 2018; Byrne et al., 2022; see Materials and Methods) and performed whole-cell patch clamp recordings in tdT+ cells to evoke AEI responses. We identified the anaxonic DA population by their characteristic 'monophasic' phase-plane plot profile (**Figure 4B**). After evoking transmitter release with transient depolarisation, we observed a prolonged inward current which was blocked by adding the GABA-A receptor antagonist gabazine at $10 \mu\text{M}$ (**Figure 4C**; AEI charge: $44.11 \pm 6.02 \text{ pC}$; $n = 31$ cells from $N = 18$ mice). These data reveal that anaxonic DA neurons display AEI responses, consistent with the co-existence of presynaptic release sites and postsynaptic targets in the dendrites of these cells.

Although the AEI response is initiated by the depolarisation of the recorded anaxonic DA neuron, it has always been assumed, but not demonstrated, that the response is the direct result of GABA released from the same cell. However, given the presence of other GABAergic cells in local glomerular circuits (Kiyokage et al., 2017; for review see Tufo et al., 2022), there are other potential indirect ways in which depolarising a DA neuron might produce a GABA response in that same cell.

DA neurons in the OB release GABA and dopamine (Borisovska et al., 2013), and D1-like and D2-like dopamine receptors are expressed in the glomerular layer (Coronas et al., 1997). When we depolarise an anaxonic DA neuron to evoke neurotransmitter release, it is therefore possible that the dopamine that is co-released with GABA activates the dopamine receptors of neighbouring neurons, evoking secondary GABA release that then drives the AEI response in the recorded cell (**Figure 4D**). To rule out this possibility, we showed that despite a non-significant trend towards a reduction in charge, the AEI response was still clearly present after the application of D1-like and D2-like receptor antagonists ($10 \mu\text{M}$ SKF 83566-hydrobromide and $10 \mu\text{M}$ sulpiride, respectively; **Figure 4E, F**; AEI charge before blockers: $25.93 \pm 4.99 \text{ pC}$, charge after blockers: $17.37 \pm 3.47 \text{ pC}$; $n = 6$ cells from $N = 4$ mice; paired t-test, $p = 0.21$). We also saw no significant alteration in AEI decay kinetics after the application of dopamine receptor antagonists (**Supplementary Figure 3**; decay constant k before blockers: $0.03 \pm 0.003 \text{ ms}^{-1}$; k after blockers: $0.03 \pm 0.006 \text{ ms}^{-1}$; $n = 6$ cells from $N = 4$ mice; Wilcoxon matched-pairs test, $p = 0.84$). These results show that the AEI response is not mediated by dopamine acting through other neurons.

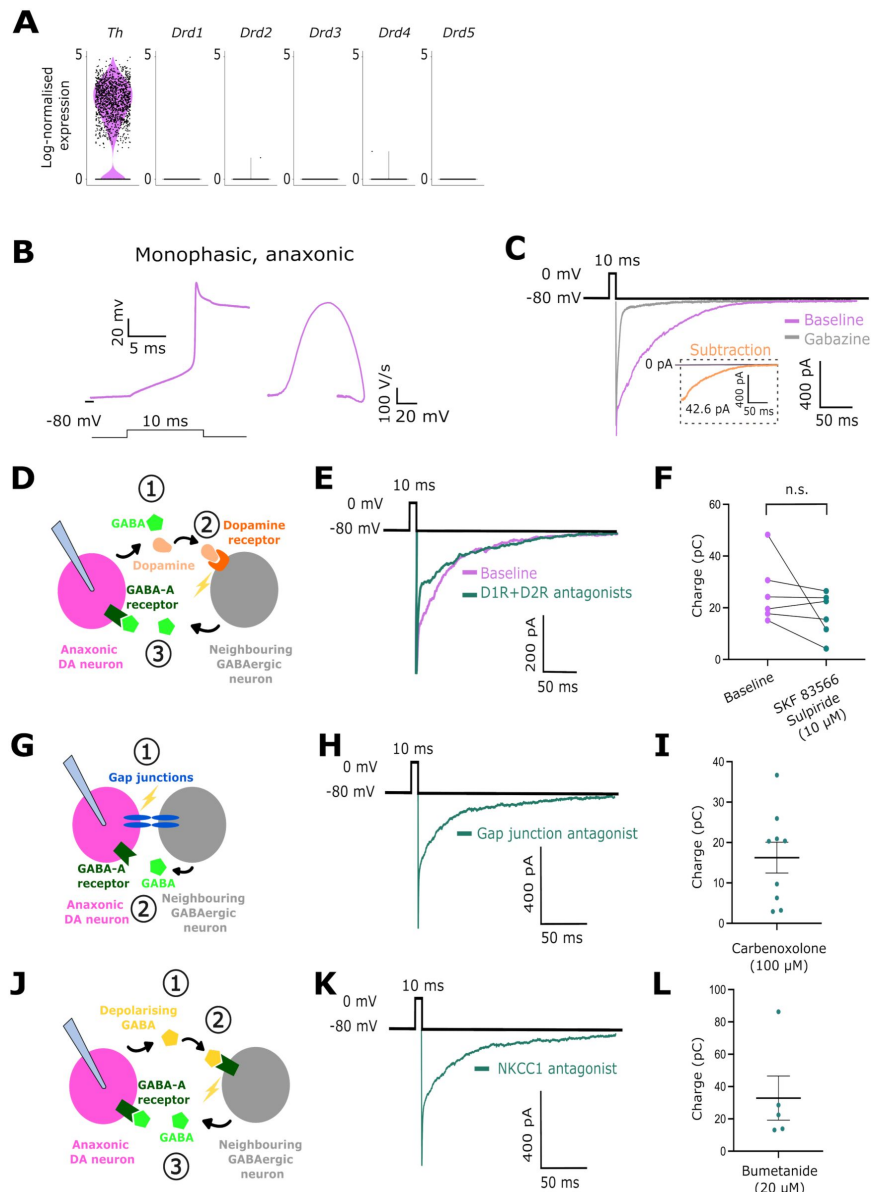


Figure 4.

Anaxonic DA neurons are capable of self-inhibition

(A) Log₂-normalised expression of *Th*, *Drd1*, *Drd2*, *Drd3*, *Drd4* and *Drd5* mRNA in OB DA cells. **(B)** Example trace of an action potential fired by a putative anaxonic DA neuron (left) and its monophasic phase-plane plot profile (right). Note the prolonged repolarisation due to Cs-based internal solution. **(C)** Example traces of an AEI response recorded before (magenta) and after (grey) the application of gabazine. The subtraction is shown in the orange inset trace. **(D)** Schematic showing the potential involvement in the AEI response of neighbouring GABAergic neurons activated by dopamine released from the patched DA cell. **(E)** Example traces of an AEI response before (purple) and after (green) applying D1-like and D2-like receptor blockers (hydrobromide and sulpiride, each at 10 μ M). **(F)** AEI charge before (purple) and after (green) applying dopamine receptor antagonists; n = 6 cells from N = 4 mice; paired t-test, p = 0.21, n.s. = non-significant. **(G)** Schematic showing the potential involvement in the AEI response of neighbouring GABAergic neurons activated via gap junctions. **(H)** Example trace of an AEI response in the presence of the gap junction blocker carbenoxolone at 100 μ M. **(I)** AEI charge in the presence of carbenoxolone at 100 μ M. Each dot shows one cell; lines show mean $\pm$ SEM; n = 9 cells from N = 4 mice. **(J)** Schematic showing the potential involvement in the AEI response of neighbouring GABAergic neurons activated by depolarising GABA released from the patched cell. **(K)** Example trace of an AEI response in the presence of the NKCC1 channel blocker bumetanide at 20 μ M. **(L)** AEI charge in the presence of bumetanide at 20 μ M. All conventions as in F; n = 5 cells from N = 3 mice.

Glomerular neurons are electrically coupled via gap junctions (Schoppa and Westbrook, 2002 [↗](#); Kosaka et al., 2005 [↗](#); Banerjee et al., 2015 [↗](#)). To exclude the possibility that the electrical activation of neighbouring, GABA-releasing neurons is mediating the AEI response (**Figure 4G** [↗](#)), we showed that AEI was still present after gap junctions were pharmacologically blocked with 100 μ M carbenoxolone (**Figure 4H, I** [↗](#); AEI charge after blockers: 16.27 ± 3.83 pC; $n = 9$ cells from $N = 4$ mice).

Finally, another source of activation of neighbouring GABA-releasing neurons could be GABA itself: GABA can have depolarising effects on glomerular layer neurons (Parsa et al., 2015 [↗](#)). To show that the GABA released by the recorded neuron is not activating neighbouring GABA-releasing cells (**Figure 4J** [↗](#)), we recorded strong AEI responses when depolarising GABA responses were prevented in the presence of the Na-K-Cl cotransporter 1 (NKCC1) blocker bumetanide used at 20 μ M (**Figure 4K, L** [↗](#); AEI charge after blockers: 32.86 ± 13.66 ; $n = 5$ cells from $N = 3$ mice). Altogether, these data robustly demonstrate that the AEI response in anaxonic DA neurons indicates true self-inhibition.

Axon-bearing DA neurons do not self-inhibit

We next wanted to assess whether axon-bearing DA neurons can self-inhibit. In light of our anatomical data showing the presence of release sites exclusively on the distal axon, we predicted that these cells do not exert synaptic inputs onto themselves. Using the same DAT-tdT mouse line to label DA neurons, we identified tdT⁺ axon-bearing DA neurons by their characteristic ‘biphasic’ action potential waveform (**Figure 5A** [↗](#), Galliano et al., 2018 [↗](#); Galliano et al., 2021; Tufo et al., 2025 [↗](#)). Consistent with axon-bearing neurons having larger soma sizes than their anaxonic counterparts (Kosaka and Kosaka, 2008 [↗](#); Galliano et al., 2018 [↗](#); **Figure 3D** [↗](#), **Supplementary Figure 2A** [↗](#)), we found that axon-bearing DA neurons exhibited significantly lower input resistance and a non-significant trend towards higher membrane capacitance compared to anaxonic DA neurons (**Supplementary Figure 4A, B** [↗](#); input resistance: 324.0 ± 31.93 M Ω for axon-bearing DA neurons, $n=11$ cells from $N=8$ mice; 683.8 ± 47.11 M Ω for anaxonic DA cells, $n=45$ cells from $N=32$ mice, $p<0.0001$, Mann-Whitney test. Membrane capacitance: 46.08 ± 4.79 pF for axon-bearing DA neurons, $n=11$ cells from $N=8$ mice; 41.57 ± 1.35 pF for anaxonic DA cells, $n=45$ cells from $N=32$ mice, $p=0.50$, Mann Whitney test). As predicted, GABA-mediated AEI currents were lacking in almost all axon-bearing DA neurons (**Figure 5B, C** [↗](#); 8/9 cells from $N = 5$ mice, = 89 %; mean $\pm$ SEM AEI charge for axon-bearing neurons: 3.8 ± 2.8 pC; $n = 9$ cells from $N = 5$ mice; anaxonic cells: 44.11 ± 6.02 pC; $n = 31$ cells from $N = 18$ mice; Mann-Whitney test, $p < 0.0001$) despite these neurons having functional GABA-A receptors as shown by the presence of prominent gabazine-sensitive spontaneous IPSCs (**Figure 5D** [↗](#); $n = 9$ axon-bearing neurons from $N = 5$ mice; $n = 31$ anaxonic cells from $N = 18$ mice). Even when we attempted to maximise AEI amplitude via a paired-pulse protocol (Bouhours et al., 2011 [↗](#)) or prolonged depolarisation, we still saw no evidence of self-inhibition in this cell subtype (**Figure 5E, F** [↗](#), **Supplementary Figure 4C, D** [↗](#); AEI charge for single response: 5.4 ± 2.61 pC; $n = 8$ cells from $N = 7$ mice; AEI charge for paired response: 1.51 ± 2.91 pC; $n = 8$ cells from $N = 7$ mice; Wilcoxon matched-pairs signed rank test, $p = 0.06$. AEI charge for 10 ms response: 0.26 ± 0.21 pC, $n = 7$ cells from $N = 6$ mice; AEI charge for 100 ms response: 7.75 ± 7.51 pC; $n = 7$ cells from $N = 6$ mice; Wilcoxon matched-pairs signed rank test, $p = 0.30$). Axon-bearing DA neurons did produce a tail current in response to the AEI protocol; however, this was insensitive to gabazine and was instead abolished by adding the voltage-gated calcium channel blocker cadmium at 200 μ M (**Figure 5G** [↗](#) and **Supplementary Figure 4E** [↗](#); average cadmium sensitive current: 15.33 ± 6.95 pC, $n=3$ cells from $N=3$ mice).

In conclusion, these findings provide both structural and functional evidence showing that anaxonic DA neurons in the OB can signal locally, even to the point of self-inhibition. In contrast, axon-bearing DA neurons can – unlike all other major cell types in this brain region – only release neurotransmitters distally from their axon.

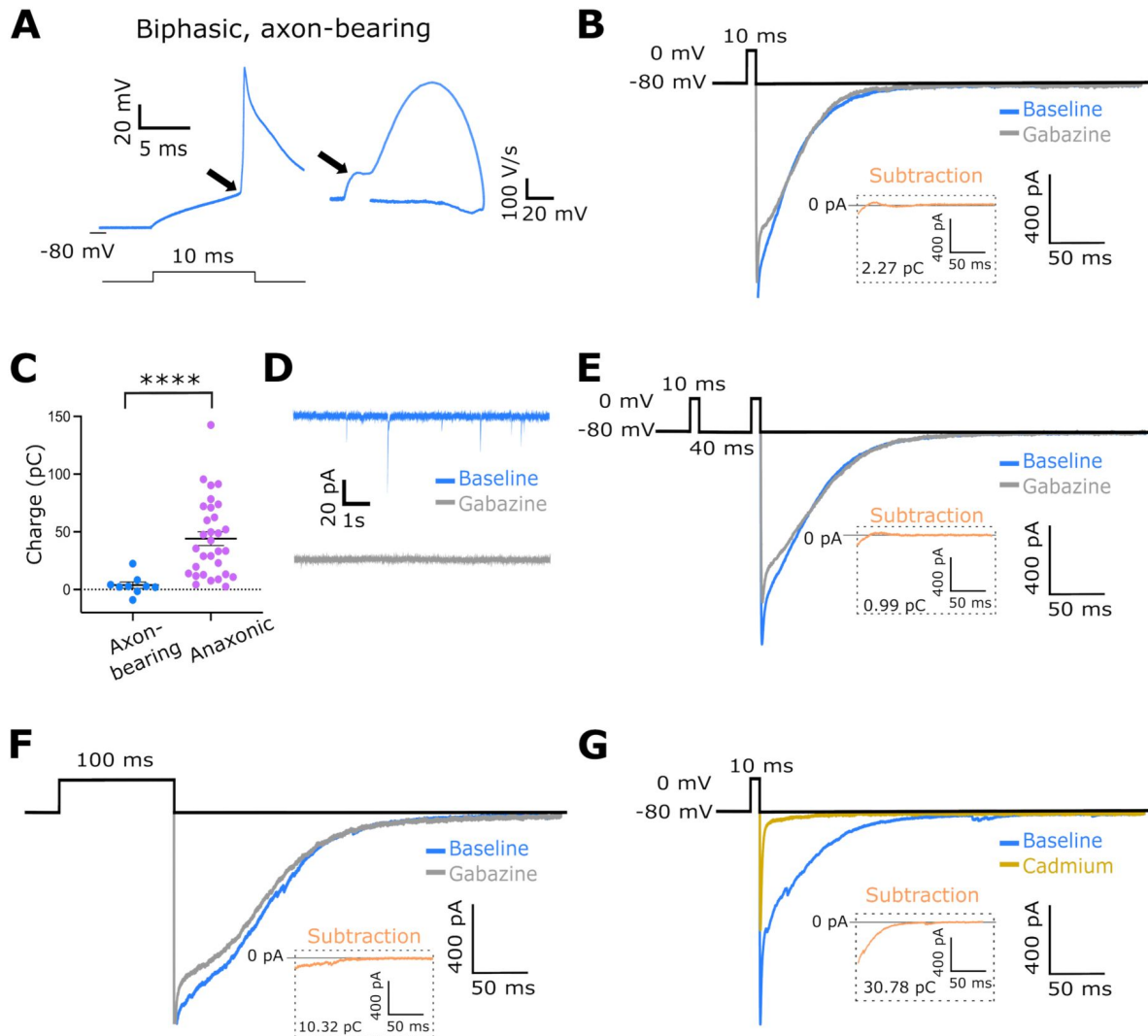


Figure 5.

Axon-bearing DA neurons do not self-inhibit

(A) Example trace of an action potential fired by a putative axon-bearing DA neuron (left) and its biphasic phase-plane plot profile (right). Arrows point to spike onset kink (left) and initial segment bump (right). Note the prolonged repolarisation due to Cs-based internal solution. **(B)** Example traces showing tail current responses in an axon-bearing DA neuron before (blue) and after (grey) the application of gabazine. The subtraction is shown in the orange inset trace. **(C)** AEI charge in axon-bearing and anaxonic DA neurons. Each dot shows one cell; lines show mean $\pm$ SEM; $n = 9$ axon-bearing cells, $n = 31$ anaxonic cells from $N = 18$ mice; Mann-Whitney test; ****, $p < 0.0001$. **(D)** Example traces showing spontaneous IPSCs in an axon-bearing DA neuron before (blue) and after (grey) the application of gabazine. **(E), (F)** Example traces showing tail current responses in axon-bearing DA neurons before (blue) and after (grey) the application of gabazine using a paired-pulse protocol **(E)** and prolonged depolarisation **(F)**. Subtractions are shown in the orange inset traces. **(G)** Example traces showing that the tail current recorded in axon-bearing DA neurons (blue) is blocked by cadmium (ocre). The subtraction is shown in the orange inset trace.

Discussion

Our results show that different polarity-defined dopaminergic (DA) subtypes in the olfactory bulb use distinct anatomical compartments for neurotransmitter release. While anaxonic DA neurons release from the dendrites, axon-bearing DA cells release nearly exclusively from the axon. These anatomical findings correlate strongly with different function: anaxonic DA neurons can self-inhibit, but this ability is absent in almost all of the axon-bearing DA population.

The link between neuronal polarity and neurotransmitter release

Our anatomical data indicate that anaxonic DA neurons have release sites on their dendrites, while the presynaptic sites of almost all the axon-bearing DA cells are located exclusively on their intermittently myelinated axons. These findings are entirely consistent with previous descriptions of release site localisation in similar OB cell types. Early ultrastructural work showed the absence of release sites in the dendrites of axon-bearing neurons in the glomerular layer, defined morphologically as ‘short-axon cells’ (Pinching and Powell, 1971 [\[1\]](#)). It was also shown that axon-bearing DA neurons located in the external plexiform layer do not have dendritic release sites (Liberia et al., 2012 [\[2\]](#)). In contrast, further ultrastructural work revealed the presence of release sites in the dendrites of small DA periglomerular neurons, presumably anaxonic (Kiyokage et al., 2017 [\[3\]](#)). Our results agree with all of these findings while specifically assigning different neurotransmitter release strategies to axon-bearing and anaxonic DA neurons.

We have shown that anaxonic DA neurons use their only available anatomical compartment, the dendrites, to release neurotransmitters. This paradigm of anaxonic structural and functional polarity is unconventional but is found in other neuron types, including amacrine cells in the retina and OB granule cells (Wu et al., 2011 [\[4\]](#); Marshak, 2016 [\[5\]](#); Nunes and Kuner, 2018 [\[6\]](#)). In fact, dendritic release is not an uncommon phenomenon in the brain and is observed even in many axon-bearing neurons. Dopaminergic neurons in the midbrain, for example, are one of the most studied examples of dendritic release (Cheramy et al., 1981 [\[7\]](#); Hikima et al., 2021 [\[8\]](#)). In the olfactory bulb, most neurons with axodendritic polarity can release from both the axon and the dendrites, including mitral cells and external tufted cells (Isaacson and Strowbridge, 1998 [\[9\]](#); Schoppa and Urban, 2003 [\[10\]](#); Hayar et al., 2004 [\[11\]](#)). In this context, axon-bearing OB DA neurons are unique: they do not have dendritic release despite their dopaminergic nature, and despite their identity as OB neurons with classic axodendritic polarity.

OB DA neurons co-release both GABA and dopamine (Wachowiak and Cohen, 1999 [\[12\]](#); Berkowicz and Trombley, 2000 [\[13\]](#); Borisovska et al., 2013 [\[14\]](#); Liu et al., 2013 [\[15\]](#); Vaaga et al., 2017 [\[16\]](#); Banerjee et al., 2015 [\[17\]](#)), so do their distinct release strategies apply to both transmitters? Our anatomical approach reliably detects GABA presynaptic sites, with a high percentage of co-localisation between synaptophysin-mRuby and vGAT (Figure 1E [\[18\]](#)). However, not all synaptophysin-mRuby puncta are vGAT-positive. This could reflect technical limitations in release site detection (Risher et al., 2015 [\[19\]](#); Heise et al., 2016 [\[20\]](#)), but also indicates that vGAT-negative spots may represent specific dopamine release sites (Borisovska et al., 2013 [\[14\]](#); Liu et al., 2013 [\[15\]](#); Vaaga et al., 2017 [\[16\]](#)). Indeed, in other brain regions synaptophysin-mRuby puncta have been used to reveal sites of dopamine release (Li et al., 2005 [\[21\]](#)). In OB DA neurons GABA and dopamine release are associated with different functions (Lyons-Warren et al., 2023 [\[22\]](#)) and the two transmitters are known not to be released from the same vesicles (Borisovska et al., 2013 [\[14\]](#)). However, this does not necessarily imply that they are released from different active zones. Addressing this question in the current study was unfortunately not possible due to notorious difficulties in staining the dopamine vesicular transporter 2 (VMAT2) due to the lack of effective commercially available antibodies (Zhang et al., 2015 [\[23\]](#); Jain et al., 2023 [\[24\]](#)). Whether anaxonic and axon-bearing DA neurons release dopamine and GABA from the same sites remains an important question for future studies.

In light of our findings, many fascinating questions about anaxonic DA neurons remain to be explored. How is dendritic release developed and maintained throughout the life of the neuron? Specific mechanisms have been identified that drive the establishment and maintenance of neuronal polarity in neurons with a classic axodendritic configuration (for reviews see [Rasband, 2010](#); [Lalli, 2014](#); [Bentley and Banker, 2016](#)). Do some of these molecular events, like cytoskeleton reorganisation and/or trafficking processes, differ in the dendrites of anaxonic neurons versus the ‘standard’ dendrites of axon-bearing neurons? Future work will answer these and further questions on the developmental and functional intricacies of unconventional neurons in the brain.

Does dendritic release equal self-inhibition?

Self-inhibition in the olfactory bulb was initially discovered in a subset of periglomerular cells ([Smith and Jahr, 2002](#)) and later described specifically in dopaminergic neurons ([Maher and Westbrook, 2008](#)). Here, we find that self-inhibition is not a universal OB DA neuron feature, but instead is an almost exclusive property of anaxonic DA cells. By ruling out potential external sources of GABA in the glomerular circuit, we have also provided the clearest evidence to date for true self-inhibition in OB periglomerular cells.

The link between dendritic release of neurotransmitters and self-inhibition exists in other brain regions as well. Neurons in the substantia nigra release dopamine that binds to auto-receptors and subsequently inhibits the neuron ([Pucak and Grace, 1994](#); [Falkenburger et al., 2001](#)). Other examples include noradrenergic neurons in the locus coeruleus and serotonergic cells in the nucleus raphe dorsalis ([Aghajanian et al., 1977](#); [Liu et al., 2005](#)). However, while these neurons have a classic axodendritic polarity and are able to release from their axons as well, anaxonic DA cells constitute a distinct model as dendrites are the solely available compartment for such function.

Finally, does dopamine released from an anaxonic OB DA neuron also modulate that same neuron? Our own data demonstrate that the auto-evoked inhibition current is purely GABAergic: it is suppressed after the application of gabazine, but unaffected after blocking dopamine receptors (**Figure 4C, E, F**). In addition, our transcriptomic data clearly show that OB DA neurons do not express dopamine receptors at all (**Figure 4A**). It is therefore unlikely there is a dopaminergic component to the self-inhibition response. Previous reports of dopamine-driven responses in OB DA neurons ([Liu, 2020](#)) may instead have occurred via well-described gap junction connectivity between DA cells and dopamine-responsive ETCs ([Liu et al., 2013](#); [Banerjee et al., 2015](#)).

Implications for information processing

Anaxonic DA neurons release neurotransmitters from their dendrites, which are confined to local areas of the glomerular layer. In contrast, axon-bearing DA cells release from their axons, which spread for long distances across the glomerular layer ([Kosaka and Kosaka, 2008](#); [Galliano et al., 2018](#)). Our findings lead to the prediction of a clear division of circuit function for very closely related cell subtypes: anaxonic DA neurons take part in local, intraglomerular inhibition, while axon-bearing DA cells participate in distal, interglomerular inhibition.

Functionally, intraglomerular inhibition acts as ‘high pass filter’ for weak odour signals detected in a glomerulus, enhancing the contrast between different olfactory stimuli ([Shao et al., 2012](#); [Gire and Schoppa, 2009](#); [Zak and Schoppa, 2021](#)). Anaxonic DA neurons are therefore expected to contribute to intraglomerular inhibition via different pathways which have been generally described for periglomerular cells. This includes the exclusively intraglomerular presynaptic inhibition of olfactory sensory neurons ([Hsia et al., 1999](#); [Ennis et al., 2001](#); [McGann et al., 2005](#); [Korshunov et al., 2017](#); [McGann 2013](#); [Vaaga et al., 2017](#)), as well as the dendrodendritic inhibition of multiple neighbouring glomerular cell types, including interneurons and projection neurons ([Murphy et al., 2005](#); [Gire and Schoppa, 2009](#)). We observed

heterogeneity in the puncta density along the dendrites of anaxonic DA neurons which, potential technical considerations notwithstanding, suggests that dendritic neurotransmitter release is not uniform across these cells. This variability may reflect differences in spatial positioning within the glomeruli, potentially aligning synaptic output with local circuit demands and individual postsynaptic partner types.

Self-inhibition would be expected to decrease the capacity of anaxonic DA neurons to release GABA over time, therefore reducing inhibitory influence within glomerular circuits. Self-inhibition could therefore control the timing of intraglomerular inhibition, allowing subsequent stimuli to activate glomerular networks. Self-inhibition could also aid the high-pass filter function of glomerular networks (Gire and Schoppa, 2009 [↗](#)). If it occurs exclusively in the context of strong DA neuron activation, it could decrease intraglomerular inhibition to promote glomerular output when input strength goes past a certain threshold. There is also the possibility that self-inhibition is more of a bug than a feature. Perhaps if anaxonic DA neurons have to release GABA from their dendrites in order to perform their intraglomerular circuit functions, the chances are high that their nearby postsynaptic receptors cannot escape some of that GABAergic influence. Further work is needed to investigate how these mechanisms influence overall intraglomerular inhibition and glomerular output.

Glomeruli in the olfactory bulb create a spatial map of odour identity, where different odours specifically activate different combinations of glomeruli (Soucy et al., 2009 [↗](#); Murthy, 2011 [↗](#)). Interglomerular inhibition acting between different glomeruli selectively enhances the signal contrast in between individual stimuli, boosting odour identification and/or discrimination (Uchida et al., 2000 [↗](#); Urban, 2002 [↗](#)). This circuit function must be exclusively undertaken by axon-bearing DA neurons, due to their unique capacity for long-range axonal release of neurotransmitters (this paper; Kosaka & Kosaka, 2008 [↗](#); Liu et al., 2013 [↗](#); Whitesell et al., 2013 [↗](#); Banerjee et al., 2015 [↗](#)).

Several key traits are shared across the entire population of OB DA neurons, including their dopaminergic and GABAergic identities and their glomerular location in the olfactory circuitry. Despite these similarities, we have found fundamental differences between OB DA subtypes in terms of their anatomy, neurotransmitter release compartments and the presence/absence of self-inhibition. These are strong foundations to suggest entirely different sensory processing functions in even very closely related interneuron sub-populations.

Materials and methods

Animals

We used mice of either sex and housed them under a 12h light-dark cycle in an environmentally controlled room with access to water and food *ad libitum*. Wild-type C57/Bl6 mice (Charles River) were used either as experimental animals, or to back-cross each generation of transgenic animals. The founders of our transgenic mouse lines – vGAT-cre (Slc32a1^{tm2(cre)Low1}/J, Jax stock 016962), DAT-cre (B6.SJL-Slc6a3^{tm1.1(cre)Bknn}/J, Jax stock 006660) and flex-tdT(B6.Cg-Gt(ROSA)26Sor^{tm9(CAG-tdTomato)Hze}/J, Jax stock 007909) were purchased from The Jackson Laboratory.

Immunohistochemistry experiments with vGAT-cre mice were performed at postnatal day (P)28. Patch-clamp electrophysiology experiments with DAT-tdT mice were performed between P21-P35. All experiments were performed in accordance with the United Kingdom Animals (Scientific Procedures) Act 1986, under the auspices of Home Office personal and project licenses held by the authors.

Viral injections

To sparsely label neuronal morphology together with presynaptic release sites, vGAT-cre mice were injected *in utero* via intraventricular injection with AAV-DJ-hSyn-FLEX-mGFP-T2A-Synaptophysin-mRuby (3×10^{12} GC/ml, AAV DJ GVVC-AAV-100, Stanford Vector Core Facility) (Beier et al., 2015 [DOI](#); Kathe et al., 2022 [DOI](#)) at embryonic day (E)14. Time-mated pregnant mice were anaesthetised with isoflurane, used at 4% for general anaesthesia induction and at 2% for anaesthesia maintenance. Injections ($\approx 1\mu\text{l}$) were performed with a glass-pulled pipette (1B150-F4, World Precision Instruments) through the uterine wall into one of the lateral ventricles of the embryos. The uterine horns were then returned into the abdominal cavity, the wall and the skin were sutured, and the embryos were allowed to continue their development *in utero* and born naturally until experimental use. Post-surgery, mice recovered in a 32°C chamber and were housed individually.

Immunohistochemistry

P28 and P33 mice were anaesthetised with an overdose of pentobarbital and then perfused with 20-25 ml of PBS with heparin (A16198, Thermo Fisher Scientific) (20 units/mL), followed by 20-25 ml of 4% paraformaldehyde (PFA; P001, TAAB Laboratories) in 3% sucrose, 60 mM PIPES, 25 mM HEPES, 5 mM EGTA, and 1 mM MgCl_2 (all purchased from Sigma-Aldrich). The olfactory bulbs were dissected and post-fixed in 4% PFA overnight (4°C), then embedded in 5% agarose and sliced at 50 μm using a vibratome (VT1000S, Leica). Free-floating slices were washed with PBS 1x and permeabilised and blocked in PBS with 5% normal goat serum (NGS, G9023, Sigma-Aldrich) and 0.25% Triton (Triton X100, Sigma-Aldrich) for 2 hr at room temperature (RT). Slices were then incubated in primary antibody solution (antibodies diluted in PBS with 5% and 0.25% Triton) for 2-5 days at 4°C. Primary antibodies, working dilutions and sources are indicated in **Table 1** [DOI](#). Slices were then washed three times for 10 minutes with PBS, before being incubated in secondary antibody solution (species-appropriate Life Technologies Alexa Fluor-conjugated; all at a 1:1000 dilution in PBS with 5% NGS and 0.25% Triton) for 2h at RT. After washing 3 times for 10 minutes with PBS, slices were mounted on glass slides (J2800AMNZ, EpreDia) with FluorSave (345789, Merck).

All immunostainings involving multiple targets were performed by incubating the primary antibodies simultaneously, except for the combination of mouse anti-TH and rat anti-MBP, where a sequential incubation was necessary to avoid cross-reactivity (first TH, then MBP).

Image acquisition, analysis, and illustration

Acquisition

All images were acquired using a laser scanning confocal microscope (Zeiss LSM 710, equipped with Zen 2010 B-SP1 software) using appropriate excitation and emission filters, a pinhole of 1 AU and 40x or 63x oil immersion objectives. Laser power and gain were set to prevent fluorescence saturation and allow acquisition with an optimal signal-to-noise ratio, and z-stacks were acquired at software-defined optimal thickness and step sizes. For puncta detection and dendritic reconstruction, stacks were acquired with 1x zoom (0.033 $\mu\text{m}/\text{pixel}$), 2048 x 2048 pixels, and in z-stacks with 0.37 μm steps. The dopaminergic identity of the cells was assessed qualitatively by taking a single plane snapshot of the TH and GFP channels that included the soma. Analysed cells were located exclusively in the OB's glomerular layer, including the border with the external plexiform layer. Axon-bearing neurons displayed at least one clear elongated TRIM46-positive region (of approximately 10 to 25 μm in length) that belonged to a GFP+ process. For co-localisation analyses, stacks were acquired with 2x zoom (0.033 $\mu\text{m}/\text{pixel}$), 2048 x 2048 pixels, and

Target	Host	Dilution	Supplier	Reference	Identifier
TH	Mouse	1:500 1:5000 1:10000 1:50000	Merck	MAB318	RRID:AB_2201528
GFP	Chicken	1:1000	Abcam	ab13970	RRID:AB_300798
TRIM46	Guinea pig	1:500	Synaptic Systems	377005	RRID:AB_2721101
vGAT	Rabbit	1:1000	Synaptic Systems	131002	RRID:AB_887871
mCherry	Rat	1:500	Thermo Fisher	M11217	RRID:AB_2536611
MBP	Rabbit	1:1000	Thermo Fisher	PA5-78397	RRID:AB_2736178
MBP	Rat	1:500	Millipore	MAB386	RRID:AB_94975
Synaptophysin	Guinea pig	1:500	Synaptic Systems	101004	RRID:AB_1210382

Table 1.

Primary antibodies

in z-stacks with 0.37 μm steps. Due to the low penetration of the vGAT antibody, and following similar studies (Heise et al., 2016 [DOI](#)), these stacks were thinner (1.1–1.2 μm total depth) to capture most of the signal and perform reliable co-localisation assessment.

Analysis

Puncta detection and dendritic tracing in 3D stacks were performed using Imaris (version 9.1.2). Individual stacks were carefully examined in three dimensions by rotating the images and adjusting the zoom to ensure accurate identification of neuronal processes belonging to specific neurons. Cells for which this could not be confirmed were excluded from the analysis. Synaptophysin-mRuby belonging to specific GFP processes were detected based on their signal overlap with GFP. To automate puncta detection, synaptophysin-mRuby and gephyrin puncta were detected by estimating a punctum size of 0.5 μm and then using the ‘Quality’ filter. Detection threshold was kept within an unbiased consistent range (1000 to 2700) that allowed adaptation to variability between staining experiments and mice. Dendritic traces were performed using the ‘Filaments’ function, and the total length and maximum length of dendrites were calculated accordingly. Puncta density was estimated by dividing the total number of puncta by the total length of dendrites.

For co-localisation analysis, the 3-dimensional centroid coordinates for the synaptophysin-mRuby puncta detected in Imaris were exported and analysed using a custom-written script in Matlab. This determined, at the y- and z-axis centroid co-ordinates of each mRuby punctum, the vGAT fluorescence intensity averaged over a 1 μm window centred on the local maximum of the mRuby distribution’s x-position. If this mRuby-centred vGAT fluorescence intensity was greater than 1 standard deviation higher than local background vGAT fluorescence (taken from a 1 μm window immediately outside the x-axis border of the mRuby punctum), the punctum was considered to be ‘co-localised.’ The overall percentage of co-localised puncta was then determined for each cell.

For soma size measurements, 3D stacks were first transformed into maximum intensity projections. We used the polygon drawing tool on Fiji (Schindelin et al., 2012 [DOI](#)) to draw the soma and obtained the area measurements using the ‘measure’ function.

Illustration

All images in the manuscript displaying examples of immunostaining were taken as high-resolution snapshots on ImarisViewer (version 10.2.0). Image stitching in **Figure 3** [DOI](#) was performed using maximum intensity projections of individual stacks that were assembled using the ImageJ plugin MosaicJ (Thevenaz and Unser, 2007 [DOI](#)). Orthogonal views in **Figure 1F** [DOI](#) were created in Fiji (Schindelin et al., 2012 [DOI](#)). Schematics, drawings, and figures layouts were designed using Inkscape (version 1.3.2).

Acute slice preparation

DAT-tdT mice at P21–35 were decapitated under isoflurane anaesthesia, and the olfactory bulb was removed and transferred into ice-cold slicing medium containing (in mM): 240 sucrose, 5 KCl, 1.25 Na_2HPO_4 , 2 MgSO_4 , 1 CaCl_2 , 26 NaHCO_3 and 10 D-Glucose (all reagents from Sigma-Aldrich), bubbled with 95% O_2 and 5% CO_2 . Horizontal slices (300 μm) of the olfactory bulb were cut using a vibratome (VT1000S, Leica and 7000 smz-2, Campden Instruments) and maintained in artificial cerebrospinal fluid (ACSF) containing (in mM): 124 NaCl, 2.5 KCl, 1.25 Na_2HPO_4 , 2 MgSO_4 , 2 CaCl_2 , 26 NaHCO_3 and 15 D-Glucose, bubbled with 95% O_2 and 5% CO_2 . Slices were kept for 30 minutes in a water bath at 34 °C and then left to recover for 30 minutes at room temperature (RT).

Acute slice electrophysiology

Individual slices were placed in a chamber mounted on a FN1 Fixed Stage Nikon microscope and held in place with a stainless-steel slice anchor. Slices were continuously superfused with ACSF heated to physiologically relevant temperature ($34 \pm 1^\circ\text{C}$) using an in-line solution heater (TC-344B, Warner Instruments). The ACSF contained, at all times, the NMDA receptor blocker DL-2-Amino-5-phosphonopentanoic acid at $50 \mu\text{M}$ (APV; A5282, Sigma-Aldrich), the AMPA receptor blocker 2,3-dioxo-6-nitro-7-sulfamoyl-benzo[f]quinoxaline at $10 \mu\text{M}$ (NBQX; N183, Sigma-Aldrich). The sodium channel blocking tetrodotoxin citrate (TTX; T-550, Alomone and 1069, Tocris) was used at $10 \mu\text{M}$ for all experiments excluding action potential recordings. Block of GABA-A receptors was achieved by the application of SR-95531 gabazine (1262, Tocris) for 5 minutes in the bath at $10 \mu\text{M}$ and confirmed by the absence of IPSCs. D1-like and D2-like receptors were blocked by applying SKF 83566 hydrobromide for 5 minutes in the bath at $10 \mu\text{M}$ (1586, Tocris) and sulpiride at $10 \mu\text{M}$ (S8010, Sigma-Aldrich), respectively. Calcium channels were blocked by applying cadmium chloride (202908, Sigma) at $200 \mu\text{M}$ for 5 minutes in the bath. The gap junction blocker carbenoxolone disodium at $100 \mu\text{M}$ (3096, Tocris) and the NKCC1 blocker bumetanide at $20 \mu\text{M}$ (3108, Tocris) were applied in the bath from the beginning and kept in the ACSF for at least 20–40 minutes. We adopted this strategy due to two key considerations: (1) carbenoxolone induced instability in the membrane properties of the neurons during the initial minutes of application, and (2) sufficient time was required for chloride transporters to reach a steady intracellular chloride concentration ($[\text{Cl}^-]$) in response to bumetanide.

Recording pipettes were pulled from borosilicate glass (1B150F-4, World Precision Instruments) with a vertical puller (PC-10, Narishige). Some were tip fire-polished with a microforge (MF-830, Narishige). All auto-evoked inhibition (AEI) recordings were performed using caesium chloride intracellular solution (in mM: 125 CsCl, 10 HEPES, 0.1 EGTA, 4 MgATP, 0.3 Na_3GTP , 10 phosphocreatine, 10 GABA (280–290 mOsm and pHed to ~ 7.3 with CsOH), and pipette resistances of 2–4 M Ω .

Auto-evoked inhibition recordings

DAT-tdT neurons were visualised with a 40x water immersion objective and tdT fluorescence was revealed by LED excitation (CoolLED pE-100) with appropriate excitation and emission filters. All recorded cells were located in the OB's glomerular layer, including the border with the external plexiform layer. To ensure specificity in targeting only dopaminergic neurons, we avoided small, dimly fluorescent neurons which are likely to be of the calretinin-expressing glomerular subtype and excluded from our data any recordings showing evidence of those calretinin-positive cells' characteristically unstable and weak spiking (Benito et al., 2018 [DOI](#); Byrne et al., 2022 [DOI](#)). Recordings were performed using a HEKA EPC 10/2 amplifier coupled to PatchMaster acquisition software (HEKA Elektronik). Signals were digitised and sampled at 20 kHz (50 μs interval sample) unless otherwise stated, and Bessel filtered at 10 kHz (filter 1) and 2.9 kHz (filter 2; active filters used in voltage clamp only).

Axon-bearing and anaxonic DA neurons were identified based on their action potential waveform. For that, cells were held at -60 mV in current clamp with 100% bridge balance. 10 ms somatic current injections of $\Delta +5 \text{ pA}$ steps were applied until threshold was reached and an action potential occurred. 10 ms duration current injections were sampled at high temporal resolution 200 kHz (5 μs sample interval) and the recordings were smoothed prior to differentiation using a 20 point (100 μs) sliding filter. The first derivative of the smoothed trace was used for dV/dt and phase plane plot analyses. Neurons were then visually determined as axon-bearing or anaxonic based on their biphasic or monophasic phase plane plot profiles, respectively (Galliano et al., 2018 [DOI](#)).

Spontaneous IPSCs were recorded as inward events at -80 mV for 1 minute while $1 \mu\text{M}$ TTX was added to the ACSF. Sodium channel block was confirmed by running an IV protocol and observing the block of the large, inward sodium current.

AEI recordings were sampled at 50 kHz ($20 \mu\text{s}$ sample interval), with slow capacitance compensated. After membrane rupture, a test series was performed in voltage-clamp mode. The average current response to a double test pulse (5 ms steps, -10 and $+10$ mV) from the holding voltage of -60 mV was used to estimate series resistance (R_s ; peak current), input resistance (R_i ; steady-state holding current at new voltage) and membrane capacitance (C_M ; area under the exponentially decaying current from peak to steady-state current). Series resistance (R_s) was assessed continuously and recordings were excluded if they exceeded 30 M Ω at any point or changed more 30% during any recording.

AEI protocols were performed in voltage-clamp from a holding potential of -80 mV, unless otherwise stated. Neurons were depolarised to 0 mV for 10 or 100 ms, before returning to -80 mV. Paired recordings were two 10 ms duration voltage steps with 50 ms inter-stimulus interval. All protocols were repeated 3 times, with 30 seconds in between each sweep, and an average response calculated. Due to the presence of passive currents, sweeps were analysed from 10 ms after the end of the voltage step for 500 ms. Gabazine and cadmium-sensitive AEI currents were measured as the difference between the pre- and the post-treatment averaged traces. The final charge was calculated as the area under the curve of the subtracted response. For AEI decay kinetics analysis, the time constant k was measured from a single exponential function fitted to the average response over the same timeframe used for charge calculation.

We confirmed no change in the AEI response over time by repeating the 10 ms voltage step protocol 3 times, with 5 minutes in between each protocol. No significant change in the charge of the AEI response was found during the experiment (**Supplementary Figure 5A** [↗](#); mean $\pm$ SEM: timepoint 1, 25 ± 7.99 pC; timepoint 2, 27.29 ± 9.57 pC; timepoint 3, 29.29 ± 11.22 pC; $n = 5$, $N = 4$ mice; one-way repeated measures ANOVA, $p = 0.44$). We also ruled out the effect of somatic depolarisation on the AEI response by holding the DA neurons either at -50 mV or -80 mV for approximately 30 seconds before the start of the recordings. Neurons were then depolarised to 0 mV for 10 ms and then brought to -80 mV. The AEI response was not different between holding at -50 mV or -80 mV (**Supplementary Figure 5B** [↗](#); mean $\pm$ SEM: -50 mV response, 25.45 ± 4.53 pC; -80 mV response, 25.3 ± 7.3 pC; $n = 5$, $N = 3$ mice, paired t-test, $p = 0.96$). These results suggest that our experimental design produces a reliable and stable response to study auto-evoked inhibition in this neuronal population.

Data analysis

For RNAseq data plotting, we made use of a whole olfactory bulb single cell RNAseq dataset (Brann et al., 2020 [↗](#)), that integrates data from juvenile and adult mouse samples, contains labels for identified clusters for the different cell types, and for visualising the expression of genes of interest in identified cell types from the olfactory bulb. Details about data processing, cluster inference and annotation can be found in the original publication. Briefly, graph-based clustering was performed using the Louvain algorithm and cluster stability evaluated using Clustree. The clusters were annotated using known gene expression markers. Here, cluster 11 corresponds to the dopaminergic neurons and contained 1453 cells. These were identified by their expression of known olfactory bulb dopaminergic neuron markers, including *Th*, *Trh*, *Pax6*, amongst others. Violin plots depicting the expression of *Th*, *Syp* (Synaptophysin), *Slc32a1* (vGAT), and the dopamine receptors *Ddr1-5*, in the cell cluster corresponding to the dopaminergic neurons, were generated with the R package Seurat. The dataset is available for download in the accompanying supplementary data, alongside the code used for generating the figures.

Data analysis, statistical analysis and plotting were carried out using GraphPad (San Diego, CA), R and custom-written Matlab (Mathworks) scripts. Data are reported as mean $\pm$ SEM unless otherwise stated. 'N' represents the number of mice and 'n' is the number of cells. Details of individual statistical tests can be found in the results section and in figure legends. Data sets were tested for normality using the D'Agostino and Pearson tests where the datasets were sufficiently large, and Shapiro-Wilk test if not. Statistical tests were determined considering the distribution of the data (normal or non-normal) and sample variance (equal or unequal). All tests were two-tailed, with $\alpha = 0.05$. Our experiments addressed questions of cellular morphology and function, and datasets comprised few neurons per animal, so comparisons were performed with cell as the unit of analysis. However, for all significant results, where false positive inflation due to pseudoreplicate data is of highest concern, we additionally performed nested analyses that took intra-animal data dependence into account (Aarts et al., 2014 [DOI](#)). In all cases, these nested tests also returned significant differences between anaxonic and axon-bearing cell types.

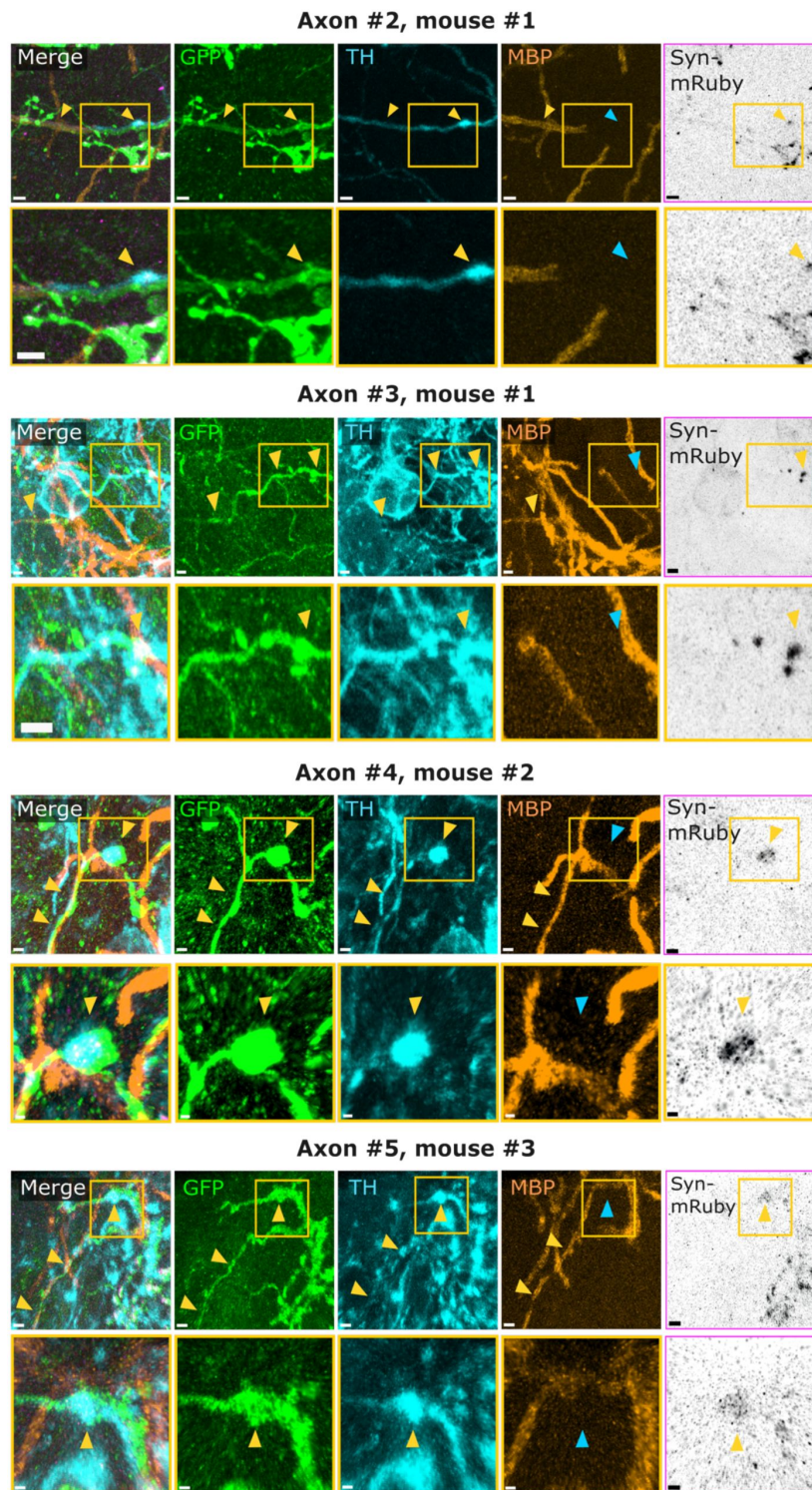
Data availability

The full raw dataset supporting this article will be openly available upon manuscript acceptance in a peer-reviewed journal at the King's College London research repository (reserved DOI: 10.18742/27731157).

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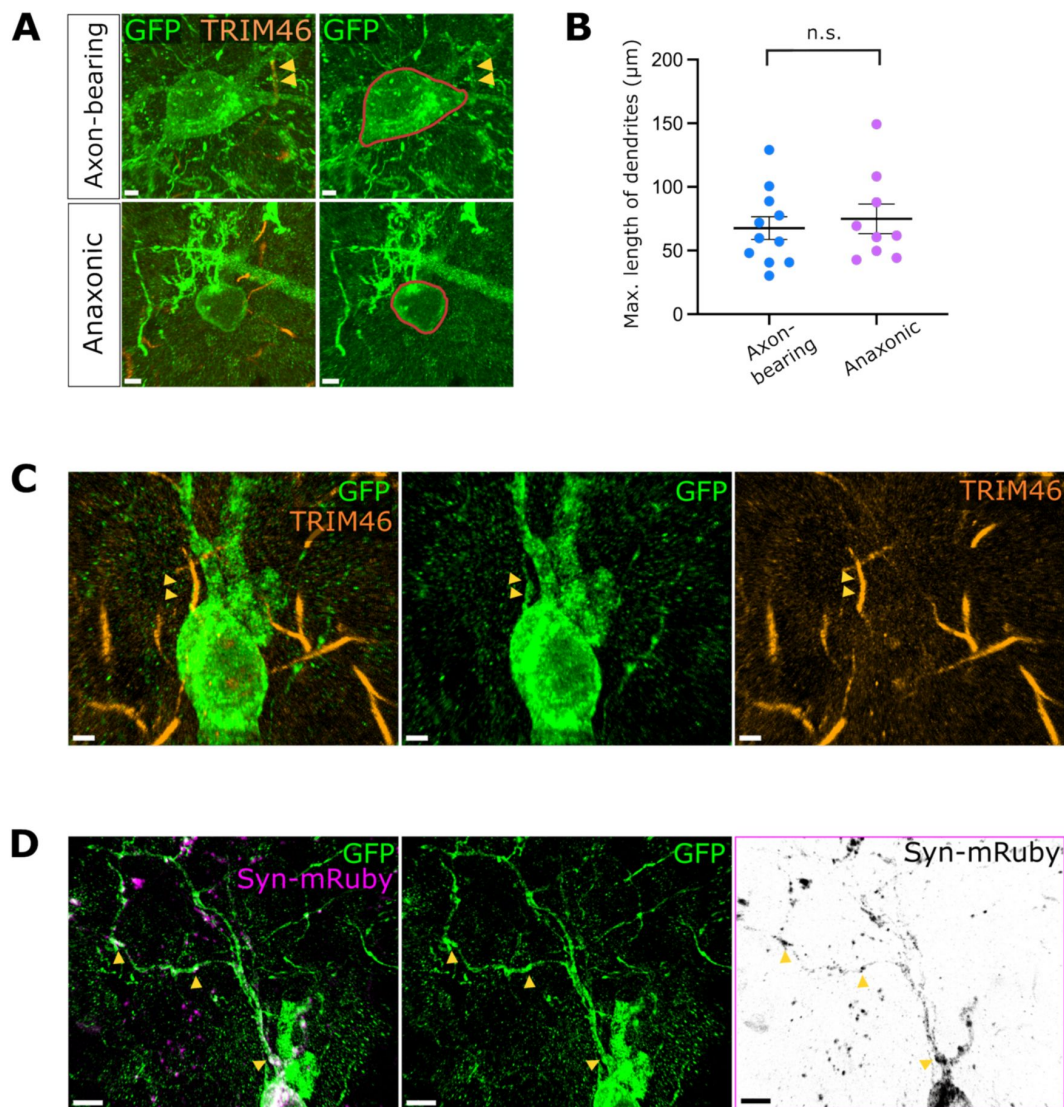
Supplementary figures



Supplementary figure 1

Example confocal images of dopaminergic axons from different mice.

Neurons are stained with GFP (green), TH (blue), MBP (orange), and Synaptophysin-mRuby (black). Yellow arrows show co-localisation between the different channels; blue arrows show lack of co-localisation. Yellow inset highlights the location of the presynaptic bouton. Images at the bottom are zoomed-in snapshots from the images at the top, and arrows point at the synaptic puncta. Scalebars for axon #2, 2 μ m and 1 μ m; for axon #3, 10 μ m and 5 μ m; for axon #4, 1 μ m and 0.5 μ m; and for axon #5, 2 μ m and 1 μ m. (related to [Figure 3](#))

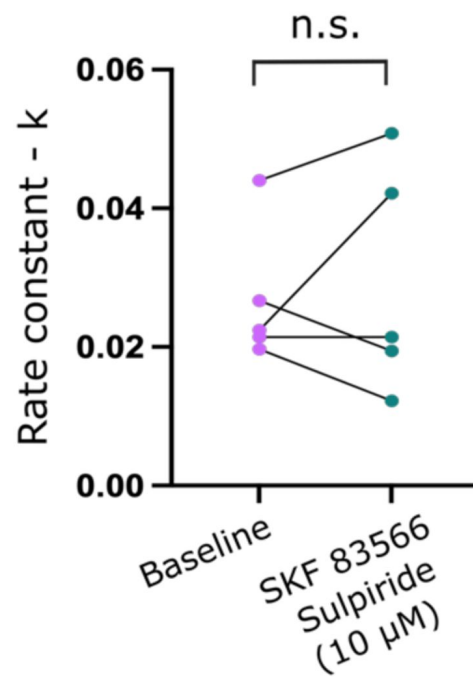


Supplementary figure 2

(A). Example confocal images showing soma size measurements in axon-bearing and anaxonic DA neurons. Axon-bearing (top) and anaxonic (bottom) cells labelled with GFP (green) and TRIM46 (orange). Yellow arrowheads point to the TRIM46-positive segment, indicating that the neuron has an axon. Red lines highlight the soma of the two neurons. Scalebars, 3 μ m.

(B) Maximum length of traced dendrites for axon-bearing (blue) and anaxonic (magenta) DA neurons. Each dot represents one cell, lines show mean $\pm$ SEM, $n = 11$ axon-bearing neurons and $n = 9$ anaxonic cells from $N = 4$ mice, Welch's t-test, $p = 0.63$, n.s. = non-significant.

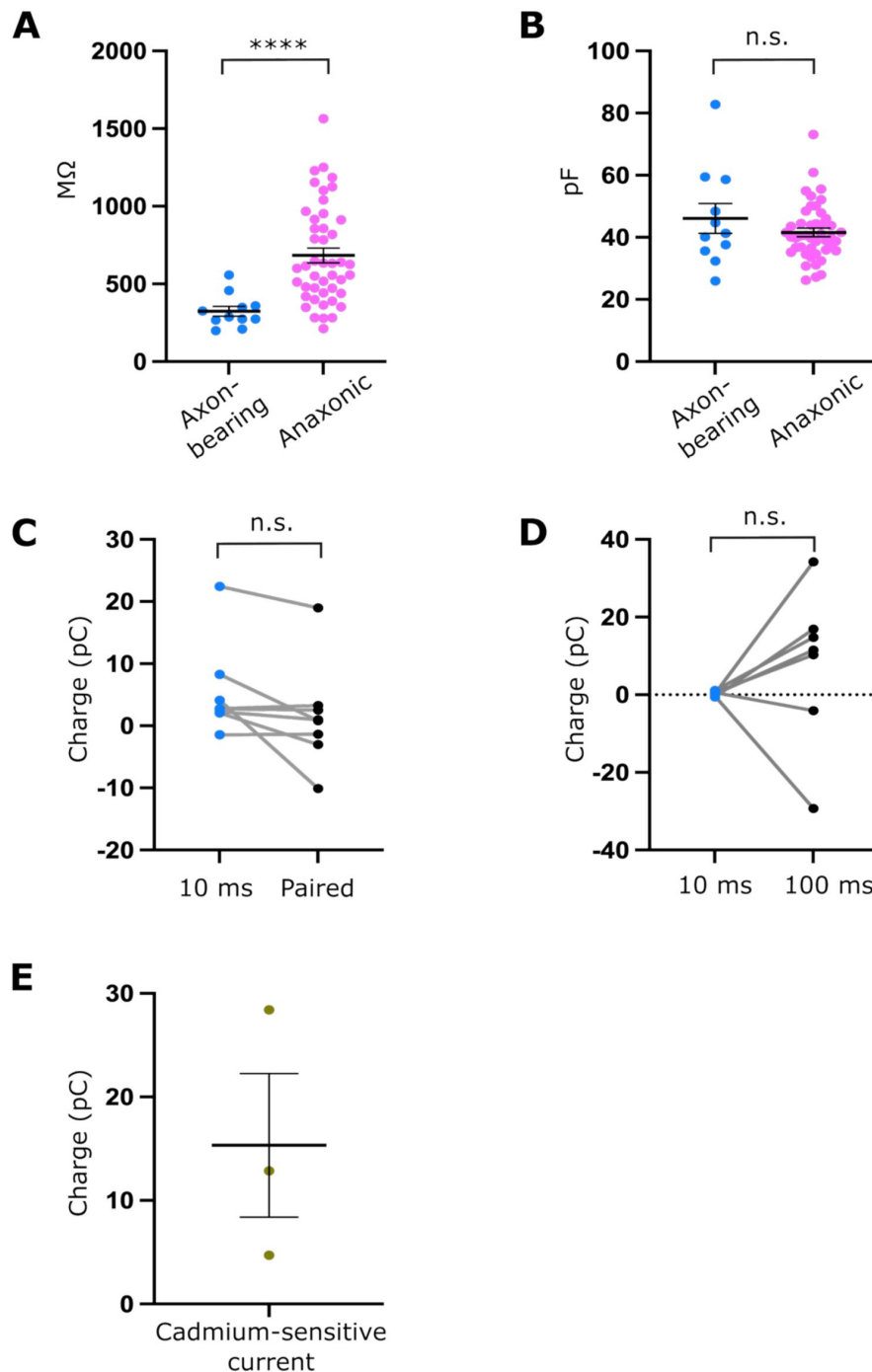
(C, D) Dendritic mRuby puncta in a strongly over-expressing axon-bearing neuron **(C)** Example confocal image with GFP (green) and TRIM46 (orange) labelling, revealing the axon-bearing identity of the neuron. Yellow arrowheads point to the GFP+/TRIM46+ co-localised zone. Scalebars, 2 μ m. **(D)** Example confocal images showing synaptophysin-mRuby label (magenta, black) in the dendrites of the axon-bearing neuron from (C) (green). The levels of synaptophysin-mRuby look dramatically higher and with a less defined puncta profile than in all anaxonic DA cells – note levels of somatic expression compared to [Figure 2B](#). Yellow arrowheads point to examples of detected puncta. Scalebars, 5 μ m. (related to [Figure 3](#))



Supplementary figure 3

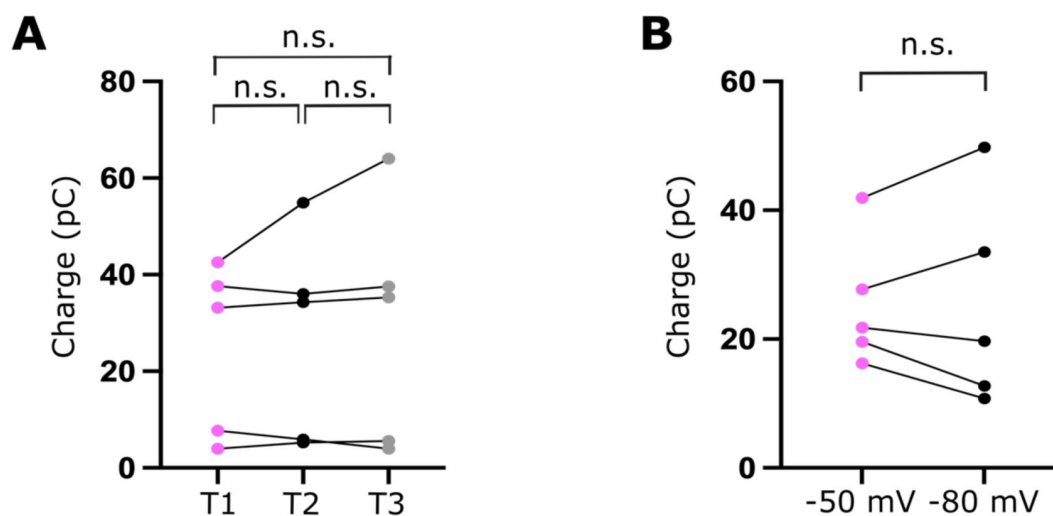
(related to Figure 4 [↗](#))

Comparison of the decay constant k of the auto-evoked inhibition response before (magenta) and after (green) the application of D1 and D2 receptor antagonists. Each dot represents one cell, $n = 6$ cells from $N = 4$ mice, Wilcoxon signed-rank test, $p = 0.84$, n.s. = non-significant.



Supplementary figure 4

(A) Input resistance in axon-bearing (blue) and anaxonic (magenta) DA neurons. Each dot represents one cell, horizontal lines show the mean $\pm$ SEM, $n = 11$ axon-bearing neurons and $n = 45$ anaxonic cells from $N = 32$ mice, ****, $p < 0.0001$, Mann-Whitney test. **(B)** Membrane capacitance in axon-bearing (blue) and anaxonic (magenta) DA neurons. All conventions as in (A), $n = 11$ axon-bearing neurons and $n = 45$ anaxonic cells from $N = 32$ mice, Mann-Whitney test, n.s. = non-significant. **(C)** Comparison of AEI charge for single (blue) and paired response (black) in axon-bearing cells. Each dot represents one cell, grey lines connect measurements from the same cell; $n = 8$ cells from $N = 7$ mice; Wilcoxon matched-pairs signed rank test, $p = 0.06$, n.s. = non-significant. **(D)** Comparison of the auto-evoked inhibition charge in axon-bearing cells following either a 10 ms or 100 ms depolarisation step. All conventions as in (C), $n = 7$ cells from $N = 6$ mice; Wilcoxon matched pairs signed rank test, $p = 0.30$, n.s. = non-significant. **(E)** Cadmium-sensitive current in axon-bearing DA neurons, $n = 3$ cells from $N = 3$ mice, all conventions as in (A). (related to [Figure 5](#))



Supplementary figure 5

(A) Comparison of the analysed auto-evoked inhibition charge after three repeats of the auto-evoked inhibition protocol. Circles represent individual cells, lines connect data from the same neuron; $n = 5$ cells from $N = 4$ mice; repeated measures one-way ANOVA, $p = 0.44$, n.s. = non-significant. (B) Comparison of the auto-evoked inhibition charge from a holding potential of either -50 mV or -80 mV. All conventions as in (A), $n = 5$ cells from $N = 3$ mice, paired t-test, $p = 0.96$, n.s. = non-significant. (related to the auto-evoked inhibition recordings section of Materials and Methods)

Additional information

Author contributions

DJB and MSG designed experiments; AD-R, DJB, YL, MC, CA and MCF performed experiments; AD-R, DJB, YL, ML and MSG analysed data, AD-R and MSG wrote the paper.

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Reviewer #1 (Public review):

Summary:

Dorrego-Rivas et al. investigated two different DA neurons and their neurotransmitter release properties in the main olfactory bulb. They found that the two different DA neurons in mostly glomerular layers have different morphologies as well as electrophysiological properties. The anaxonic DA neurons are able to self-inhibit but the axon-bearing ones are not. The findings are interesting and important to increase the understanding both of the synaptic transmissions in the main olfactory bulb and the DA neuron diversity. However, there are some major questions that the authors need to address to support their conclusions.

(1) It is known that there are two types of DA neurons in the glomerular layer with different diameters and capacitances (Kosaka and Kosaka, 2008; Pignatelli et al., 2005; Angela Pignatelli and Ottorino Belluzzi, 2017). In this manuscript, the authors need to articulate better which layer the imaging and ephys recordings took place, all glomerular layers or with an exception. Meanwhile, they have to report the electrophysiological properties of their recordings, including capacitances, input resistance, etc.

(2) It is understandable that recording the DA neurons in the glomerular layer is not easy. However, the authors still need to increase their n's and repeat the experiments at least three times to make their conclusion more solid. For example (but not limited to), Fig 3B, n=2 cells from 1 mouse. Fig.4G, the recording only has 3 cells.

(3) The statistics also use pseudoreplicates. It might be better to present the biology replicates, too.

(4) In Figure 4D, the authors report the values in the manuscript. It is recommended to make a bar graph to be more intuitive.

(5) In Figure 4F and G, although the data with three cells suggest no phenotype, the kinetics looked different. So, the authors might need to explore that aside from increasing the n.

(6) Similarly, for Figure 4I and J, L and M, it is better to present and analyze it like F and G, instead of showing only the after-antagonist effect.

Comments on revisions:

In the rebuttal, the authors argued that it had been extremely hard to obtain recordings stable enough for before-and-after effects on the same cell. Alternatively, they could perform the before-and-after comparison on different cells.

<https://doi.org/10.7554/eLife.105271.2.sa2>

Reviewer #2 (Public review):

Summary:

This study provides novel insights into the neurotransmitter release mechanisms employed by two distinct subclasses of dopaminergic neurons in the olfactory bulb (OB). The findings suggest that anaxonic neurons primarily release neurotransmitters through their dendrites, whereas axon-bearing neurons predominantly release neurotransmitters via their axons. Furthermore, the study reveals that anaxonic neurons exhibit self-inhibitory behavior, indicating that closely related neuronal subclasses may possess specialized roles in sensory processing.

Strengths:

This study introduces a novel and significant concept, demonstrating that two closely related neuron subclasses can exhibit distinct patterns of neurotransmitter release. Therefore, this finding establishes a valuable framework for future investigations into the functional diversity of neuronal subclasses and their contributions to sensory processing. Furthermore, these findings offer fundamental insights into the neural circuitry of the olfactory bulb, enhancing our understanding of sensory information processing within this critical brain region.

Weaknesses:

The reliance on synaptophysin-based presynaptic structures raises minor concerns about whether these structures represent functional synapses.

Comments on revisions:

Most of the concerns have been addressed by the authors, and there are no further comments about this manuscript.

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

This Reviewer was positive about the study, stating 'The findings are interesting and important to increase the understanding both of the synaptic transmissions in the main olfactory bulb and the DA neuron diversity.' They provided a number of helpful suggestions for improving the paper, which we have incorporated as follows:

(1) It is known that there are two types of DA neurons in the glomerular layer with different diameters and capacitances (Kosaka and Kosaka, 2008; Pignatelli et al., 2005; Angela Pignatelli and Ottorino Belluzzi, 2017). In this manuscript, the authors need to articulate better which layer the imaging and ephys recordings took place, all glomerular layers or with an exception. Meanwhile, they have to report the electrophysiological properties of their recordings, including capacitances, input resistance, etc.

We thank the Reviewer for this clarification. Indeed, the two dopaminergic cell types we study here correspond directly to the subtypes previously identified based on cell size. Our previous work showed that axon-bearing OB DA neurons have significantly larger somas than their anaxonic neighbours (Galliano et al. 2018), and we replicate this important result in the present study (Figure 3D). In terms of electrophysiological correlates of cell size, we now provide full details of passive membrane properties in the new Supplementary Figure 4, as requested. Axon-bearing DA neurons have significantly lower input resistance and show a non-significant trend towards higher cell capacitance. Both features are entirely consistent with the larger soma size in this subtype. We apologise for the oversight in not fully describing previous categorisations of OB DA neurons, and have now added this information and the appropriate citations to the Introduction (lines 56 to 59 of the revised manuscript).

In terms of cell location, all cells in this study were located in the OB glomerular layer. We sampled the entire glomerular layer in all experiments, including the glomerular/EPL border where the majority of axon-bearing neurons are located (Galliano et al. 2018). This is now clarified in the Materials and Methods section (lines 535 to 537 and 614 to 616 of the revised manuscript).

(2) It is understandable that recording the DA neurons in the glomerular layer is not easy. However, the authors still need to increase their n's and repeat the experiments at least three times to make their conclusion more solid. For example (but not limited to), Fig 3B, n=2 cells from 1 mouse. Fig.4G, the recording only has 3 cells.

Despite the acknowledged difficulty of these experiments, we have now added substantial extra data to the study as requested. We have increased the number of cells and animals to further support the following findings:

Fig 3B: we now have n=5 cells from N=3 mice. We have created a new Supplementary Figure 1 to show all the examples.

Figure 4G: we now have n=6 cells from N=4 mice.

Figure 5G: we now have n=3 cells from N=3 mice.

The new data now provide stronger support for our original conclusions. In the case of auto-evoked inhibition after the application of D1 and D2 receptor antagonists, a nonsignificant trend in the data suggests that, while dopamine is clearly not necessary for the response, it may play a small part in its strength. We have now included this consideration in the Results section (lines 256 to 264 of the revised manuscript).

(3) The statistics also use pseudoreplicates. It might be better to present the biology replicates, too.

Indeed, in a study focused on the structural and functional properties of individual neurons, we performed all comparisons with cell as the unit of analysis. This did often (though not always) involve obtaining multiple data points from individual mice, but in these low-throughput experiments n was never hugely bigger than N . The potential impact of pseudoreplicates and their associated within-animal correlations was therefore low. We checked this in response to the Reviewer's comment by running parallel nested analyses for all comparisons that returned significant differences in the original submission. These are the cases in which we would be most concerned about potential false positive results arising from intra-animal correlations, which nested tests specifically take into account (Aarts et al., 2013). In every instance we found that the nested tests also reported significant differences between anaxonic and axonbearing cell types, thus fully validating our original statistical approach. We now report this in the relevant section of the Materials and Methods (lines 686 to 691 of the revised manuscript).

(4) In Figure 4D, the authors report the values in the manuscript. It is recommended to make a bar graph to be more intuitive.

This plot does already exist in the original manuscript. We originally describe these data to support the observation that an auto-evoked inhibition effect exists in anaxonic neurons (corresponding to now lines 240 to 245 of the revised manuscript). We then show them visually in their entirety when we compare them to the lack of response in axon-bearing neurons, depicted in Figure 5C. We still believe that this order of presentation is most appropriate for the flow of information in the paper, so have maintained it in our revised submission.

(5) In Figure 4F and G, although the data with three cells suggest no phenotype, the kinetics looked different. So, the authors might need to explore that aside from increasing the n .

We thank the Reviewer for this suggestion. To quantify potential changes in the autoevoked inhibition response kinetics, we fitted single exponential functions and compared changes in the rate constant (k ; Methods, lines 650 to 652 of the revised manuscript). Overall, we observed no consistent or significant change in rate constant values after adding DA receptor antagonists. This finding is now reported in the Results section (lines 260 to 263 of the revised manuscript) and shown in a new Supplementary Figure 3.

(6) Similarly, for Figure 4I and J, L and M, it is better to present and analyze it like F and G, instead of showing only the after-antagonist effect.

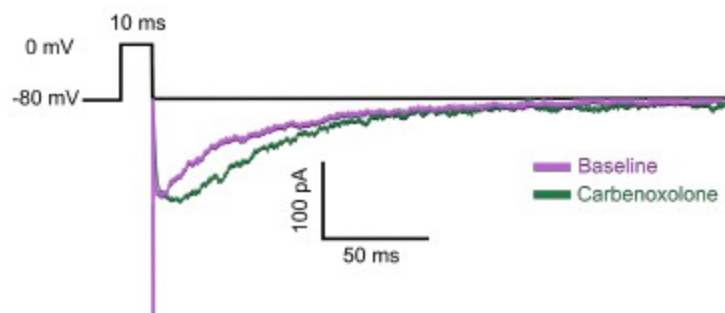
We agree that the ideal scenario would have been to perform the experiments in Figure 4J and 4M the same way as those in Figure 4G, with a before vs after comparison. Unfortunately, however, this was not practically possible.

When attempting to apply carbenoxelone to already-patched cells, we found that this drug highly disrupted the overall health and stability of our recordings immediately after its

application. This is consistent with previous reports of similar issues with this compound (e.g. Connors 2012, *Epilepsy Currents*; Tovar et al., 2009, *Journal of Neurophysiology*). After many such attempts, the total yield of this experiment was one single cell from one animal. Even so, as shown in the traces below, we were able to show that the auto-evoked inhibition response was not eliminated in this specific case:

Author response image 1.

Traces of an AEI response recorded before (magenta) and after (green) the application of carbenoxolone (n=1 cell from N=1 mouse).



In light of these issues, we instead followed published protocols in applying the carbenoxolone directly in the bath without prior recording for 20 minutes (following Samailova et al., 2003, *Journal of Neurochemistry*) and ran the protocol after that time. Given that our main question was to ask whether gap junctions were strictly necessary for the presence of any auto-evoked inhibition response, our positive findings in these experiments still allowed us to draw clear conclusions.

In contrast, the issue with the NKCC1 antagonist bumetanide was time. As acknowledged by this Reviewer, obtaining and maintaining high-quality patch recordings from OB DA neurons is technically challenging. Bumetanide is a slow-acting drug when used to modify neuronal chloride concentrations, because in addition to the time it takes to reach the neurons and effectively block NKCC1, the intracellular levels of chloride subsequently change slowly. Studies using this drug in slice physiology experiments typically use an incubation time of at least 20 minutes (e.g. Huberfeld et al., 2007, *Journal of Neuroscience*), which was incompatible with productive data collection in OB DA neurons. Again, after many unsuccessful efforts, we were forced instead to include bumetanide in the bath without prior recording for 20-30 minutes. As with the carbenoxolone experiment, our goal here was to establish whether autoevoked inhibition was in any way retained in the presence of this drug, so our positive result again allowed us to draw clear conclusions.

Reviewer #1 (Recommendations for the authors):

(1) I suggest the authors reconsider the terminology. For example, they use "strikingly" in their title. The manuscript reported two different transmitter release strategies but not the mechanisms, and the word "strikingly" is not professional, either.

We appreciate the Reviewer's attention to clarity and tone in the manuscript title, and have nevertheless decided to retain the original wording. The almost all-or-nothing differences between closely related cell types shown in structural and functional properties here (Figures 3F & 5C) are pronounced, extremely clear and easily spotted – all properties appropriate for the word 'striking.' In addition, we note that the use of this term is not at all unprofessional, with a PubMed search for 'strikingly' in the title of publications returning over 200 hits.

(2) Similarly, almost all confocal scopes are 3D because images can be taken at stacks. So "3D confocal" is misleading.

We understand that this is misleading. We have now replaced the sentence 'Example snapshot of a 3D confocal stack of...' by 'Example confocal images of...' in all the figure legends that apply.

(3) It is recommended to present the data in bar graphs with data dots instead of showing the numbers in the manuscript directly.

We agree entirely, and now present data plots for all comparisons reported in the study (Supplementary Figures 2, 4 and 5).

Reviewer #2 (Recommendations for the authors):

(1) Several experiments report notably small sample sizes, such as in Figures 3B and 5G, where data from only 2 cells derived from 1-2 mice are presented. Figures 4E-G also report the experimental result only from 3 cells derived from 3 mice. To enhance the statistical robustness and reliability of the findings, these experiments should be replicated with larger sample sizes.

As per our response to Reviewer 1's comment #2 above, and to directly address the concern that some evidence was 'incomplete', we have now added significant extra data and analysis to this revised submission (Figures 4 and 5; and Supplementary Figure 1). We believe that this has further enhanced the robustness and reliability of our findings, as requested.

(2) The authors utilize vGAT-Cre for Figures 1-3 and DAT-tdTomato for Figures 4-5, raising concerns about consistency in targeting the same population of dopaminergic neurons. It remains unclear whether all OB DA neurons express vGAT and release GABA. Clarification and additional evidence are needed to confirm whether the same neuronal population was studied across these experiments.

Although we indeed used different mouse lines to investigate structural and functional aspects of transmitter release, we can be very confident that both approaches allowed us to study the same two distinct DA cell types being compared in this paper. Existing data to support this position are already clear and strong, so in this revision we have focused on the Reviewer's suggestion to clarify the approaches we chose.

First, it is well characterised that in mouse and many other species all OB DA neurons are also GABAergic. This has been demonstrated comprehensively at the level of neurochemical identity and in terms of dopamine/GABA co-release, and is true across both small-soma/anaxonic and large-soma/axon-bearing subclasses (Kosaka & Kosaka 2008; 2016; Maher & Westbrook 2008; Borisovska et al., 2013; Vaaga et al., 2016; Liu et al. 2013). To specifically confirm vGAT expression, we have also now provided additional single-cell RNAseq data and immunohistochemical label in a revised Figure 1 (see also Panzanelli et al., 2007, now referenced in the paper, who confirmed endogenous vGAT colocalisation in TH-positive OB neurons). Most importantly, by using vGAT-cre mice here we were able to obtain sufficient numbers of both anaxonic and axon-bearing DA neurons among the vGAT-cre-expressing OB population. We could unambiguously identify these cells as dopaminergic because of their expression of TH protein which, due to the absence of noradrenergic neurons in the OB, is a specific and comprehensive marker for dopaminergic cells in this brain region (Hokfelt et al., 1975; Rosser et al., 1986; Kosaka & Kosaka 2016). Crucially, both axon-bearing and anaxonic OB DA subtypes strongly express TH (Galliano et al., 2018, 2021). We have now added

additional text to the relevant Results section (lines 99 to 108 of the revised manuscript) to clarify these reasons for studying vGAT-cre mice here.

We were also able to clearly identify and sample both subtypes of OB DA neuron using DAT-tdT mice. Our previous published work has thoroughly characterised this exact mouse line at the exact ages studied in the present paper (Galliano et al., 2018; Byrne et al., 2022). We know that DAT-tdT mice provide rather specific label for TH-expressing OB DA neurons (75% co-localisation; Byrne et al., 2022), but most importantly we know which non-DA neurons are labelled in this mouse line and how to avoid them. All nonTH-expressing but tdT-positive cells in juvenile DAT-tdT mice are small, dimly fluorescent and weakly spiking neurons of the calretinin-expressing glomerular subtype (Byrne et al., 2022). These cells are easily detected during physiological recordings, and were excluded from our study here. This information is now provided in the relevant Methods section (lines 616 to 619 of the revised manuscript, also referenced in lines 236 to 240 of the results section), and we apologise for its previous omission. Finally, we have shown both structurally and functionally that both axon-bearing and anaxonic OB DA subtypes are labelled in DAT-tdT mice (Galliano et al., 2018, Tufo et al., 2025; present study). Overall, these additional clarifications firmly establish that the same neuronal populations were indeed studied across our experiments.

(3) The low TH+ signal in Figure 1D raises questions regarding the successful targeting of OB DA neurons. Further validation, such as additional staining, is required to ensure that the targeted neurons are accurately identified.

As noted in our response to the previous comment, TH is a specific marker for dopaminergic neurons in the mouse OB, and is widely used for this purpose. Labelling for TH in our tissue is extremely reliable, and in fact gives such strong signal that we were forced to reduce the primary antibody concentration to 1:50,000 to prevent bleedthrough into other acquisition channels. Even at this concentration it was extremely straightforward to unambiguously identify TH-positive cells based on somatic immunofluorescence. We recognise, however, that the original example image in Figure 1D was not sufficiently clear, and have now provided a new example which illustrates the TH-based identification of these cells much more effectively.

(4) Estimating the total number of dopaminergic neurons in the olfactory bulb, along with the relative proportions of anaxonic and axon-bearing neuron subtypes, would provide valuable context for the study. Presenting such data is crucial to underscore the biological significance of the findings.

This information has already been well characterised in previous studies. Total dopaminergic cell number in the OB is ~90,000 (Maclean & Shipley, 1988; Panzanelli et al., 2007; Parrish-Aungst et al., 2007). In terms of proportions, anaxonic neurons make up the vast majority of these cells, with axon-bearing neurons representing only ~2.5% of all OB dopaminergic neurons at P28 (Galliano et al., 2018). Of course, the relatively low number of the axon-bearing subtype does not preclude its having a potentially large influence on glomerular networks and sensory processing, as demonstrated by multiple studies showing the functional effects of inter-glomerular inhibition (Kosaka & Kosaka, 2008; Liu et al., 2013; Whitesell et al., 2013; Banerjee et al., 2015). This information has now been added to the Introduction (line 47 and lines 59 to 62 of the revised manuscript).

(5) The authors report that in-utero injection was performed based on the premise that the two subclasses of dopaminergic neurons in the olfactory bulb are generated during embryonic development. However, it remains unclear whether in-utero injection is essential for distinguishing between these two subclasses. While the manuscript references a relevant study, the explanation provided is insufficient. A more detailed

justification for employing in-utero injection would enhance the manuscript's clarity and methodological rigor.

We apologise for the lack of clarity in explaining the approach. In utero injection is not absolutely essential for distinguishing between the two subclasses, but it does have two major advantages. 1) Because infection happens before cells migrate to their final positions, it produces sparse labelling which permits later unambiguous identification of individual cells' processes; and 2) Because both subclasses are generated embryonically (compared to the postnatal production of only anaxonic DA neurons), it allows effective targeting of both cell types. We have now expanded the relevant section of the Results to explain the rationale for our approach in more detail (lines 109 to 116 of the revised manuscript).

(6) In Figures 1A and 4A, it appears that data from previously published studies were utilized to illustrate the differential mRNA expression in dopaminergic neurons of the olfactory bulb. However, the Methods section and the manuscript lack a detailed description of how these dopaminergic neurons were classified or analyzed. Given that these figures contribute to the primary dataset, providing additional explanation and context is essential to ensure clarity of the findings.

We apologise for the lack of clarity. We have now extended the part of the methods referring to the RNAseq data analysis (lines 666 to 678 of the revised manuscript).

(7) In Figure 2C, anaxonic dopamine neurons display considerable variability in the number of neurotransmitter release sites, with some neurons exhibiting sparse sites while others exhibit numerous sites. The authors should address the potential biological or methodological reasons for this variability and discuss its significance.

We thank the Reviewer for highlighting this feature of our data. We have now outlined potential methodological reasons for the variability, whilst also acknowledging that it is consistent with previous reports of presynaptic site distributions in these cells (Kiyokage et al., 2017; Results, lines 169 to 172 of the revised manuscript). We have also added a brief discussion of the potential biological significance (Discussion, lines 446 to 450).

(8) In the images used to differentiate anaxonic and axon-bearing neurons, the soma, axons, and dendrites are intermixed, making it difficult to distinguish structures specific to each subclass. Employing subclass-specific labeling or sparse labeling techniques could enhance clarity and accuracy in identifying these structures.

Distinguishing these structures is indeed difficult, and was the main reason we used viral label to produce sparse labelling (see response to comment #5 above). In all cases we were extremely careful, including cells only when we could be absolutely certain of their anaxonic or axon-bearing identity, and could also be certain of the continuity of all processes. Crucially, while the 2D representations we show in our figures may suggest a degree of intermixing, we performed all analyses on 3D image stacks, significantly improving our ability to accurately assign structures to individual cells. We have now added extra descriptions of this approach in the relevant Methods section (lines 546 to 548 of the revised manuscript).

(9) In Figure 3, the soma area and synaptophysin puncta density are compared between axon-bearing and anaxonic neurons. However, the figure only presents representative images of axon-bearing neurons. To ensure a fair and accurate comparison, representative images of both neuron subtypes should be included.

The original figures did include example images of puncta density (or lack of puncta) in both cell types (Figure 2B and Figure 3E). For soma area, we have now included representative

images of axon-bearing and anaxonic neurons with an indication of soma area measurement in a new Supplementary Figure 2A.

(10) In Figure 4B, the authors state that gephyrin and synaptophysin puncta are in 'very close proximity.' However, it is unclear whether this proximity is sufficient to suggest the possibility of self-inhibition. Quantifying the distance between gephyrin and synaptophysin puncta would provide critical evidence to support this claim. Additionally, analyzing the distribution and proportion of gephyrin/synaptophysin pairs in close proximity would offer further clarity and strengthen the interpretation of these findings.

We thank the Reviewer for raising this issue. We entirely agree that the example image previously shown did not constitute sufficient evidence to claim either close proximity of gephyrin and synaptophysin puncta, nor the possibility of self-inhibition. We are not in a position to perform a full quantitative analysis of these spatial distributions, nor do we think this is necessary given previous direct evidence for auto-evoked inhibition in OB dopaminergic cells (Smith and Jahr, 2002; Murphy et al., 2005; Maher and Westbrook, 2008; Borisovska et al., 2013) and our own demonstration of this phenomenon in anaxonic neurons (Figure 4). We have therefore removed the image and the reference to it in the text.

(11) In Figures 4J and 4M, the effects of the drugs are presented without a direct comparison to the control group (baseline control?). Including these baseline control data is essential to provide a clear context for interpreting the drug effects and to validate the conclusions drawn from these experiments.

We appreciate the Reviewer's attention to this important point. As this concern was also raised by Reviewer 1 (their point #6), we have provided a detailed response fully addressing it in our replies to Reviewer 1 above.

(12) In Lines 342-344, the authors claim that VMAT2 staining is notoriously difficult. However, several studies (e.g., Weihe et al., 2006; Cliburn et al., 2017) have successfully utilized VMAT2 staining. Moreover, Zhang et al., 2015 - a reference cited by the authors - demonstrates that a specific VMAT2 antibody effectively detects VMAT2. Providing evidence of VMAT2 expression in OB DA neurons would substantiate the claim that these neurons are GABA-co-releasing DA neurons and strengthen the study's conclusions.

As noted in response to this Reviewer's comment #2 above, there is clear published evidence that OB DA neurons are GABA- and dopamine-releasing cells. These cells are also known to express VMAT2 (Cave et al., 2010; Borisovska et al., 2013; Vergaña-Vera et al., 2015). We do not therefore believe that additional evidence of VMAT2 expression is necessary to strengthen our study's conclusions. We did make every effort to label VMAT2-positive release sites in our neurons, but unfortunately all commercially available antibodies were ineffective. The successful staining highlighted by the Reviewer was either performed in the context of virally driven overexpression (Zhang et al., 2015) or was obtained using custom-produced antibodies (Weihe et al., 2006; Cliburn et al., 2017). We have now modified the Discussion text to provide more clarification of these points (lines 393 to 395 of the revised manuscript).

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